विद्या परं दैवतम्
IIM
VISAKHAPATNAM

Logistics Management

2025-27 batch

(Session 1-2)



Logistics can spell the difference between success and failure in business. James L Heskett

Dr. Aalok Kumar

Course Housekeeping

Evaluation Component	Type (Individual/ Group)	Weightage	Tentative timing – Session# when it becomes due
1. Simulation	Individual	20%	Session 12-13
2. Case reflection report and discussion	Group	20%	Session 4 and session 17
3. Mid-term Exam	Individual	20%	After 10 sessions
4. End Term Exam.	Individual	40%	After 20 Sessions

Syllabus: Attached in Course pack

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Prescribed Textbook: BUSINESS LOGISTICS/SUPPLY CHAIN MANAGEMENT, Ballou and Shrivastava 5TH EDITION, ISBN: 978-8131705841, Pearson Publisher

The Historical Perspective

- Management techniques and philosophies developed by the military were adopted by the commercial sector
  - Logistics was the enabler and frequently the determining factor in success of
    - Colonization
    - The industrial revolution
    - Urbanization
    - Centralization of commerce and power
    - Increasing consumerism

Modern Perspective:

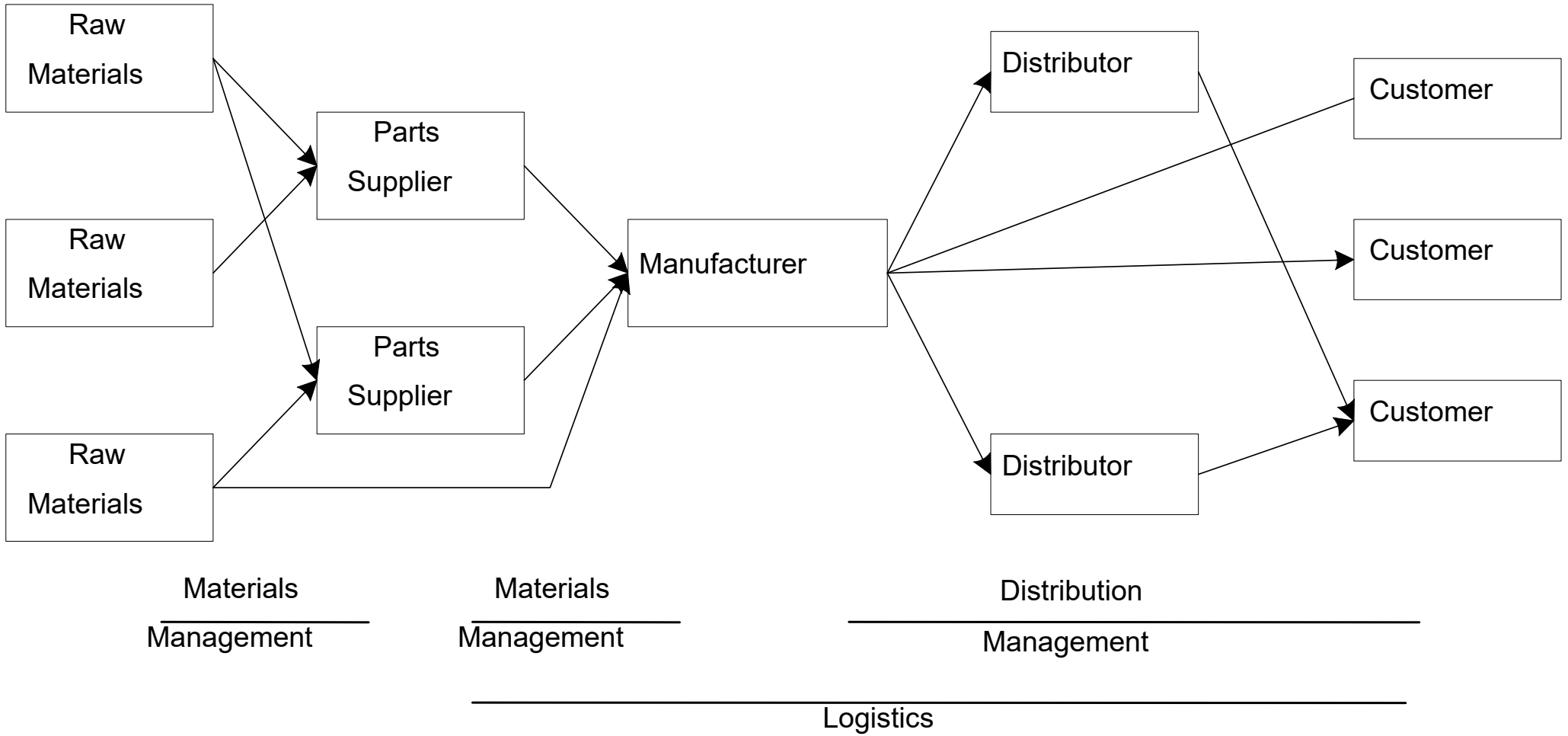
- As we study logistics in this course we will find that, at its heart, ***logistics addresses issues of customer satisfaction***
- The systematic management of the various activities required to move benefits from their point of production to the customer

Meaning of Logistics Management

- Logistics - Logistics is that part of the supply chain process that plans, implements, and controls the efficient, effective forward and reverse flow and storage of goods, services, and related information between the point of origin and the point of consumption in order to meet customers' requirements. *(CLM)*

- Supply Chain Management - the systemic, strategic coordination of the traditional (modern) business functions and the tactics across these business functions within a particular company and across businesses within the supply chain for the purposes of improving the long-term performance of the individual companies and the supply chain as a whole. *(CLM)*

The Modern Logistics System

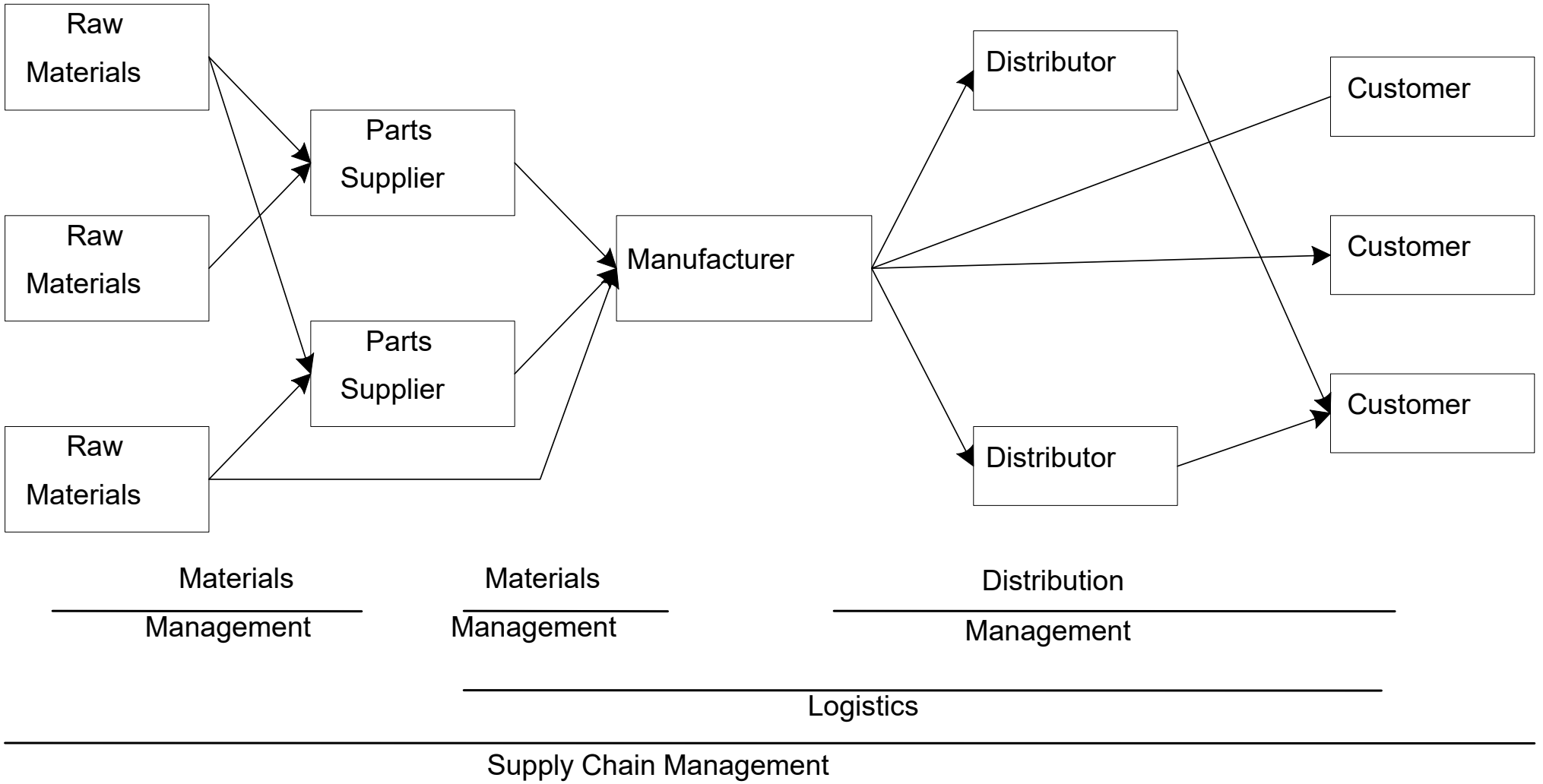


Components of Logistics Systems

- Customer service
- Inventory management
- Transportation
- Storage and materials handling
- Packaging
- Information processing
- Demand forecasting
- Purchasing
- Facility location
- Reverse Logistics
- Warranty support

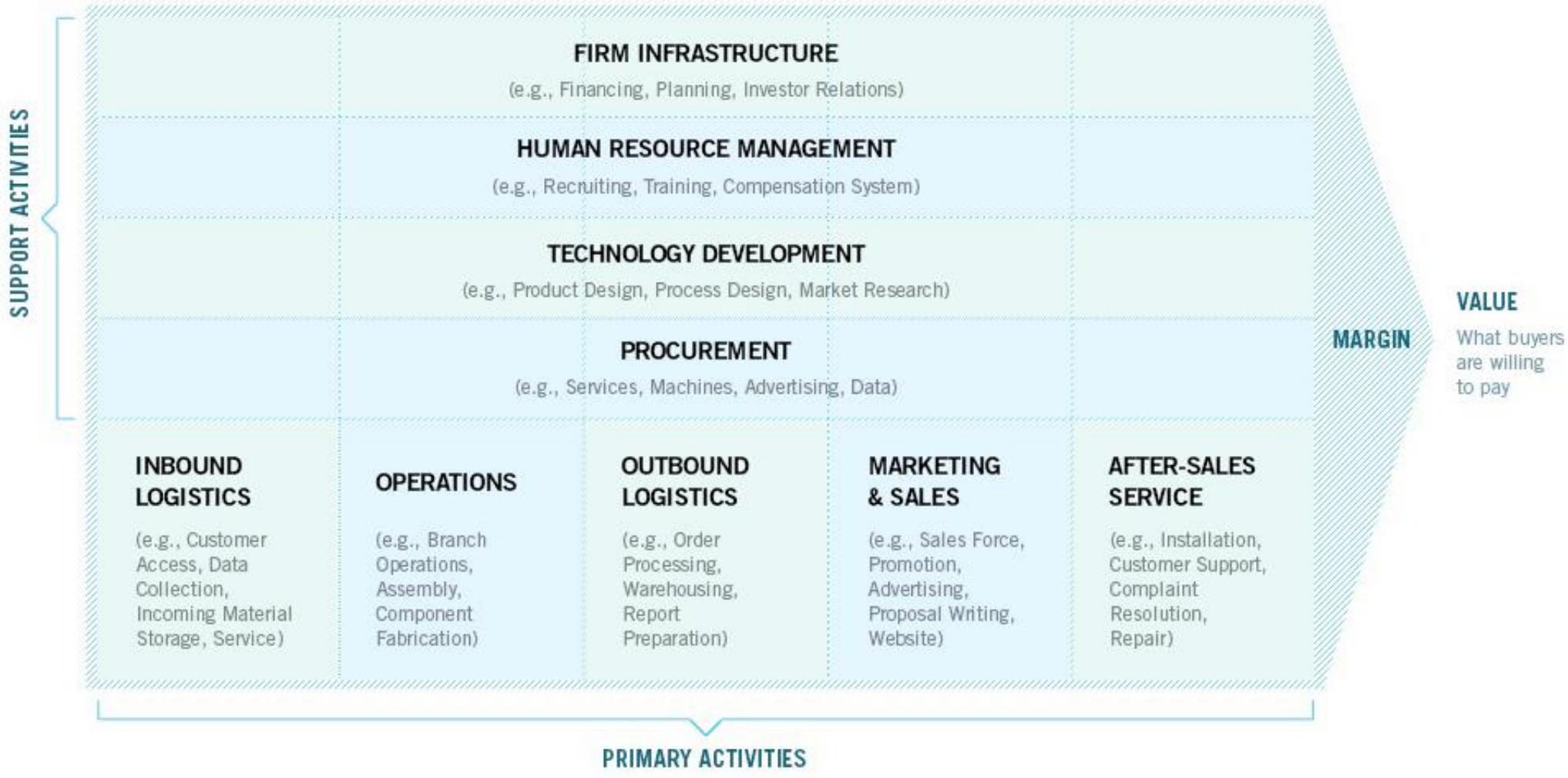
Many of these functions have traditionally been stand-alone roles within the organization….**BUT** they are all related because of their ability to impact ***customer satisfaction***.

Extended Logistics Management



Logistics in the Organization

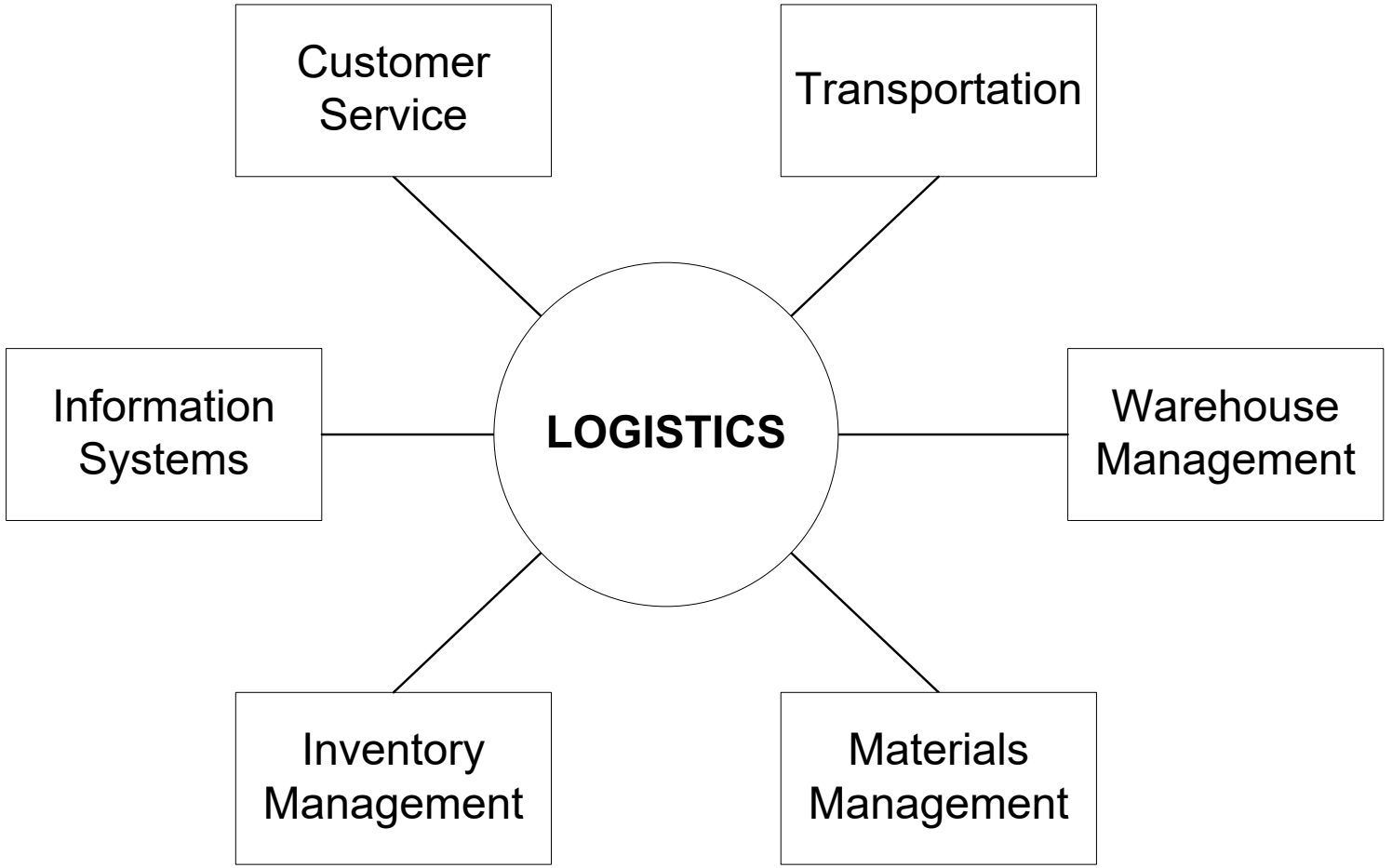
Michael Porter's concept of the value chain



Logistics as a Competitive Edge

- Logistics is important strategically and tactically in order to gain competitive advantage
- Business is increasingly aware that well-managed logistics systems can provide the organization with a sustainable competitive advantage
- Lower labor cost
- Natural resources
- Large captive markets
- Technical advantage/expertise

The Logistics System



How LM is different from SCM

- **Supply Chain Management**

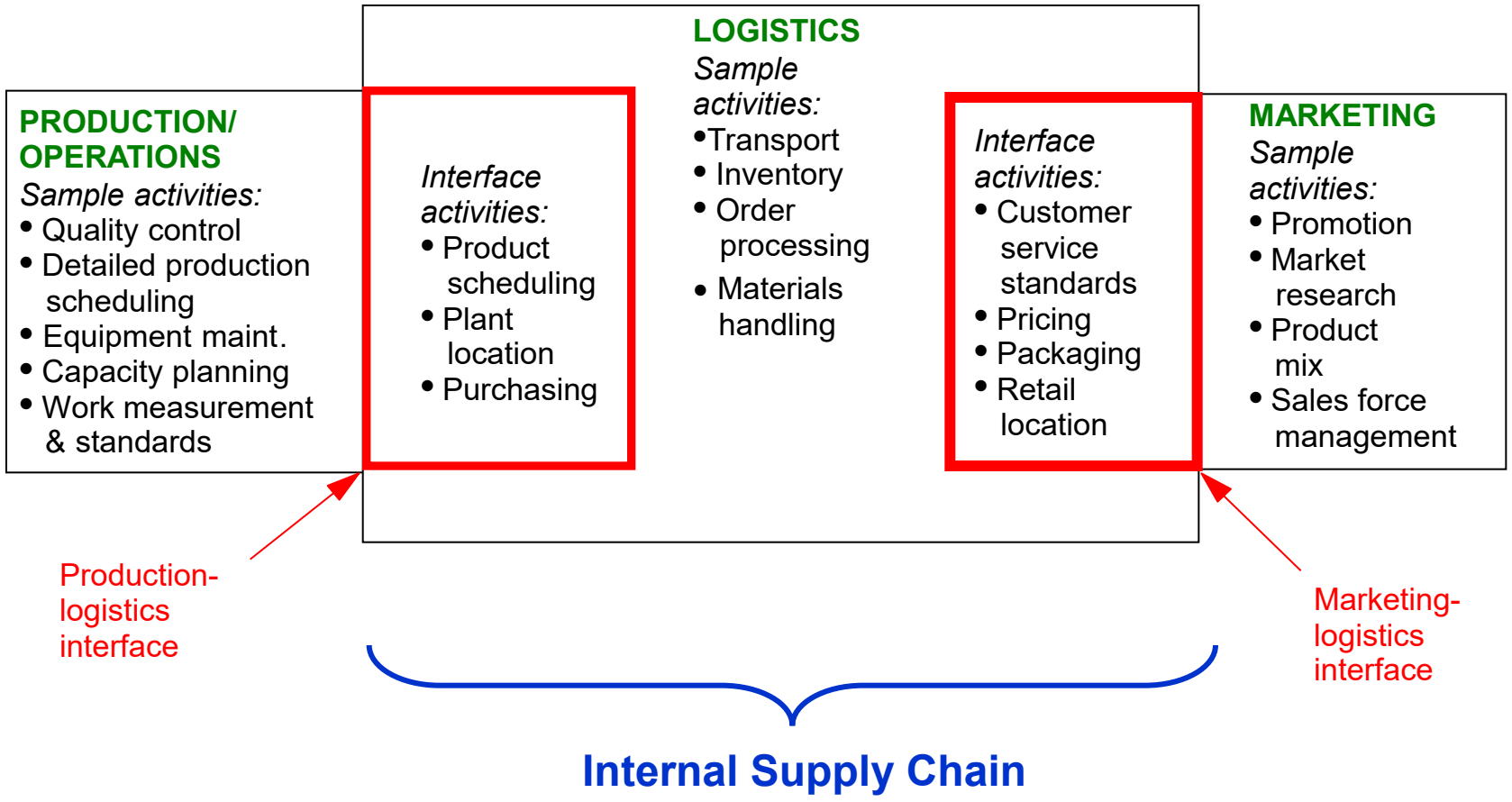
1. SCM encompasses a broader perspective and encompasses all activities involved in the sourcing, procurement, production, and distribution of goods.
2. Focus on entire network
3. Coordination and integrated system
4. Strong focus on competitive advantage

- …

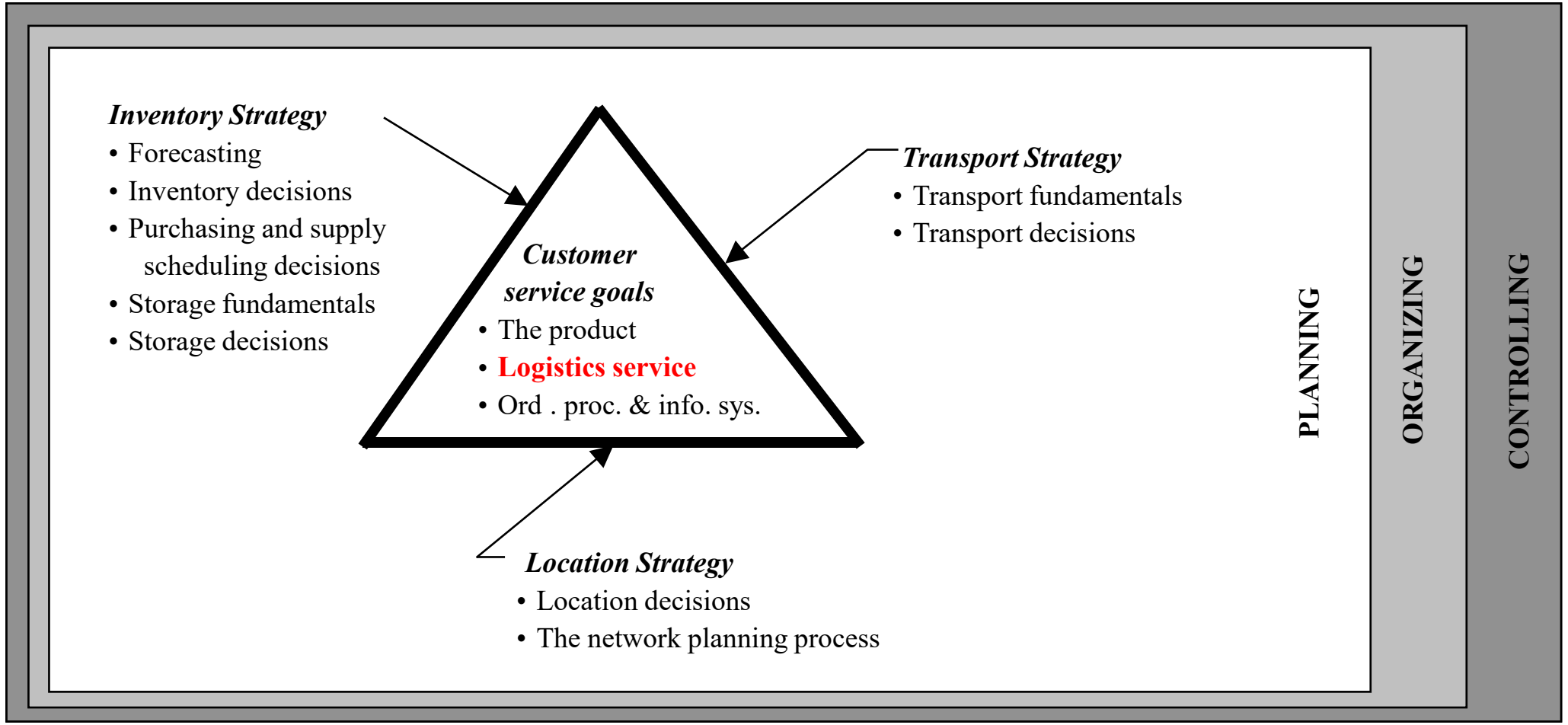
Logistics Management

1. Operational function of SCM
2. Focus on cost and time-efficient delivery
3. Coordination and integrated system
4. Strong focus on customer satisfaction
5. …..

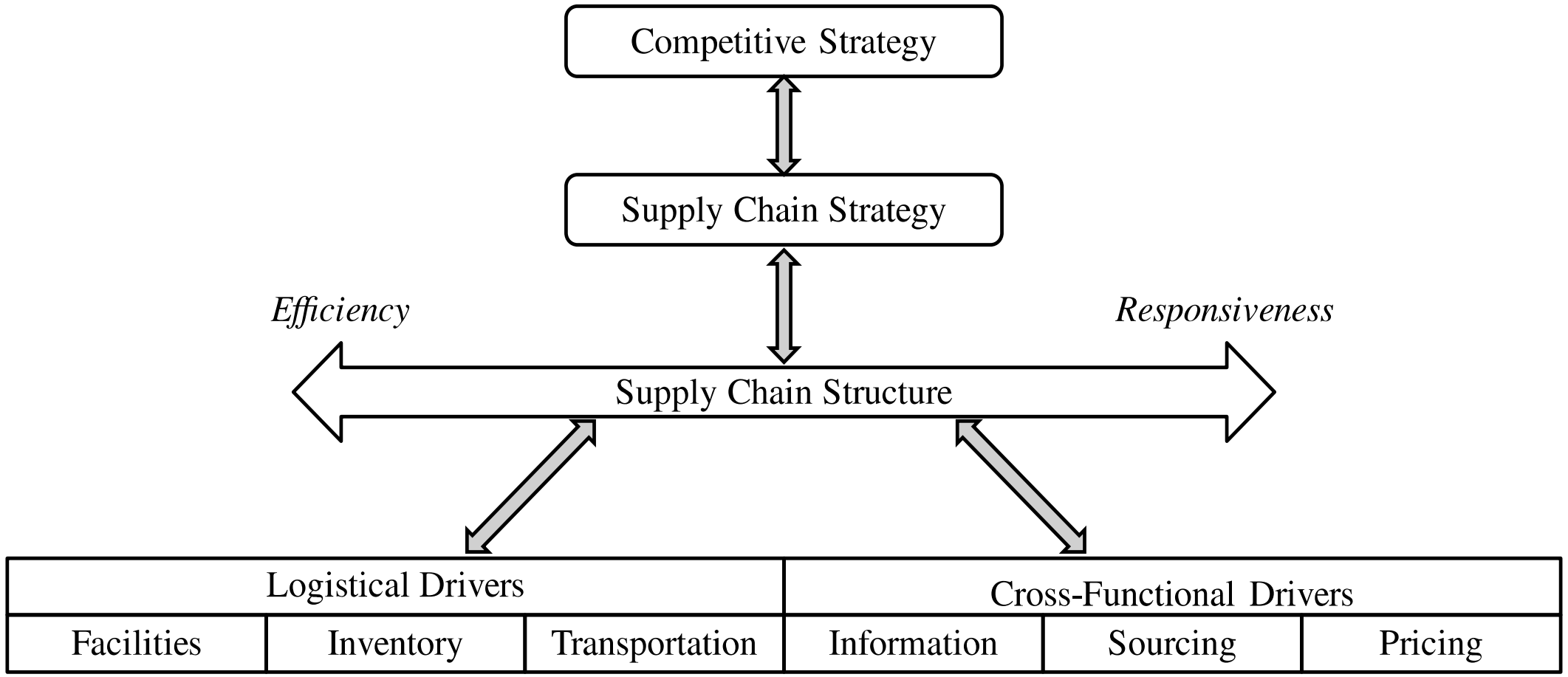
Relationship of Logistics to Market and Production



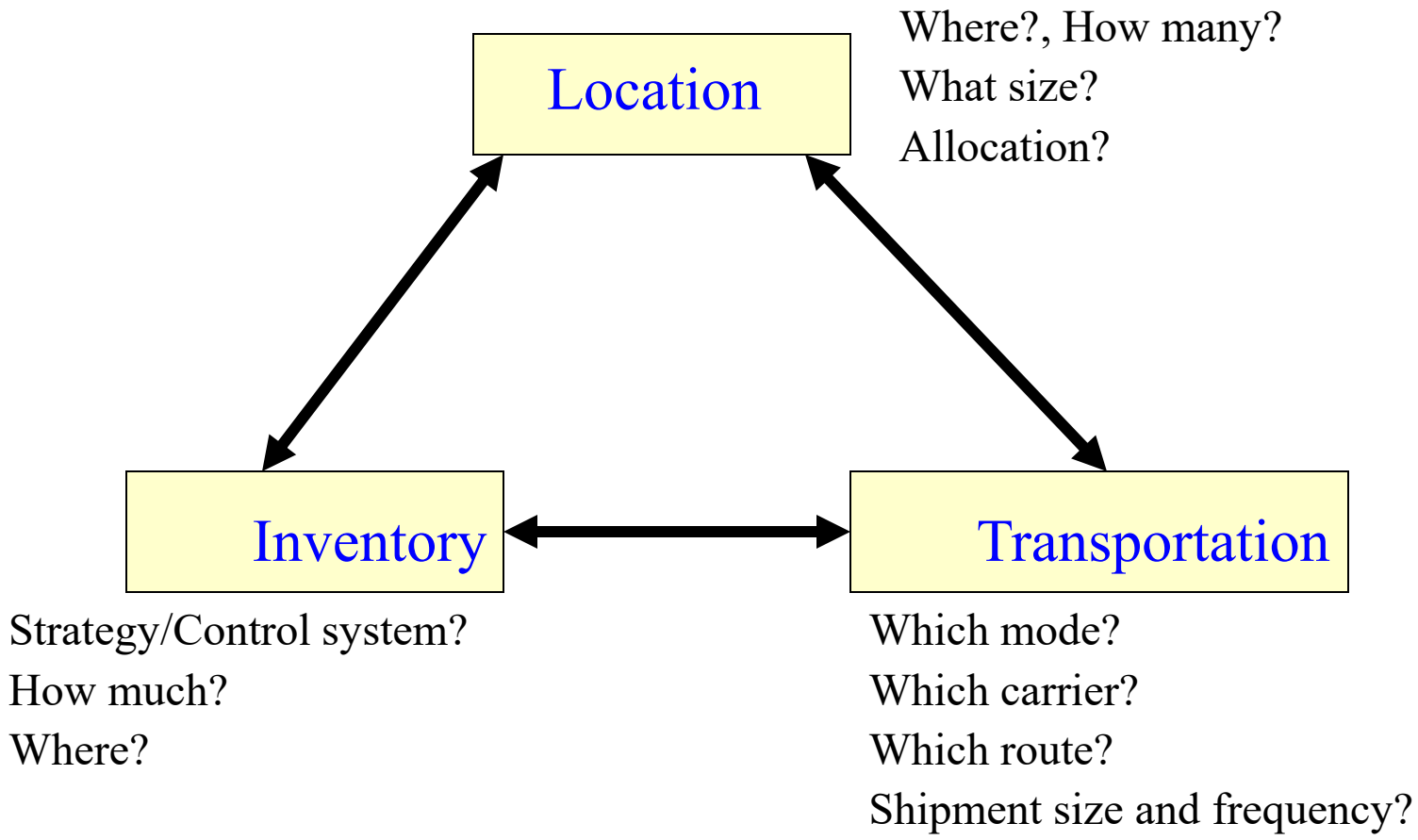
Customer Service in Planning Triangle



Logistical drivers linking with SC



Decision Areas



Strategic, Tactical, and Operational Decision Making

Decision area	*Strategic*	*Tactical*	*Operational*
Transportation	Mode selection	Seasonal equip-ment leasing	Dispatching
Inventories	Location, Control policies	Safety stock levels	Order filling
Order processing	Order entry, transmittal, and processing system design		Processing orders, Filling back orders
Purchasing	Development of supplier-buyer relations	Contracting, Forward buying	Expediting
Warehousing	Handling equipment selection, Layout design	Space utilization	Order picking and restocking
Facility location	Number, size, and location of warehouses		

Practical application of transportation costing

- To be competitive in the marketplace, businesses must focus on the speed of deliveries and price.
- The transport cost consist of Phantom Cost and Absorption cost for planning transport costing.
- Phantom cost is a shipping charge imposed upon a customer in excess of the true shipping cost incurred by the seller. This is also known as phantom freight.
- Absorption cost is a geographic pricing strategy in which a seller absorbs a part of the shipping cost in delivering the goods in order to capture the business.

Single zone pricing scheme

- Single zone pricing refers to charging the same price to all customers regardless of their geographical locations.
- Those customers whose actual shipping costs are higher than the single price will gain positive perception about the seller.
- Those customers, whose actual shipping costs are lower than the single price may gain negative perception about the seller as they realize an overcharged situation.
- Single zone pricing is practiced when sellers do not have enough pricing power to charge a different price to each buyer.

Multiple zone pricing

- In this pricing model actual shipping costs and charged shipping costs are very close so that neither sellers nor buyers have to absorb any part of shipping costs.
- An example of multizone pricing approach is priority Mail delivery by the US Postal Service (USPS).
- The USPS charges different amounts for the same Priority Mail based on zip code.

Transport zoning problem

Logistics Company A is located in Topeka, Kansas. This company ships custom-made ergonomic chairs from its manufacturing plan in Topeka, KS to its customers across the United States.

Company A adopted the single zone pricing approach for its business. Company A charges a fixed shipping cost of $49 for standard shipping per custom-made chair to any customers anywhere in the United States.

Recently, the company calculated the actual cost of shipping per chair in three different zones for their customers. Table shows the actual cost of shipping for zone A, zone B, and zone C. The table also shows the price difference for different types of shipping methods.

Transport zoning problem

- If a customer from Oklahoma City, Oklahoma orders two custom-made chairs using standard shipping from Company A, what is the phantom cost for this customer (if any) or what is the freight absorption cost for Company A (if any)?
- What would be the phantom cost for the same order, but if the customer is located in Columbus, Ohio?
- Is there any freight absorption cost for Company A for this new scenario?
- In which zone or zones does Company A need to absorb shipping costs?

Shipping method	Zone A (actual cost)	Zone B (actual cost)	Zone C (actual cost)
Standard shipping: 5–7 days	$37.25	$49.00	$65.75
Expedited shipping: 3–5 days	$55.45	$75.50	$99.00
One-day shipping: 1 business day	$110.50	$134.00	$175.00


Zone C
Zone B
Zone A
Regina
Winnipeg
Québec
Seattle
Olympia
Washington
Montana
Helena
North Dakota
Bismarck
Maine
Augusta
Ottawa
Montréal
Salem
Oregon
Idaho
Boise
South Dakota
Pierre
St. Paul
Minnesota
Wisconsin
Michigan
Toronto
Montpelier
Albany
New York
Mass.
Wyoming
Madison
Lansing
Detroit
Chicago
Pennsylvania
Philadelphia
Harrisburg
Iowa
Nebraska
Cheyenne
Des Moines
Ohio
Columbus
Indiana
Illinois
Lincoln
Salt Lake City
Carson City
San Francisco
Utah
Nevada
Denver
Colorado
Topeka
Springfield
Indianapolis
West Virginia
Richmond
Charleston
Virginia
St. Louis
Jefferson City
Frankfort
California
Las Vegas
Kansas
Missouri
Kentucky
Raleigh
Nashville
North Carolina
Tennessee
Los Angeles
Arizona
Santa Fe
Oklahoma
Memphis
Columbia
Oklahoma City
Arkansas
Atlanta
South Carolina
Phoenix
New Mexico
Little Rock
Mississippi
Georgia
Montgomery
Dallas
Jackson
Louisiana
Alabama
Texas
Tallahassee
Austin
Baton Rouge
Florida
Houston
Miami

Solution

- Standard shipping charge: $49.00 per chair
- Actual cost incurred to ship to Oklahoma City: $37.25 per chair
- Customer was charged 2*49= $98.00 for two chairs, while Company A had to pay 2*37.25=74.50 for two chairs.
- In this particular case, Company A charged more than the actual cost of shipping. Hence, the customer from Oklahoma City will see a phantom cost of (98-74.50= $23.50).
- In the second scenario, the customer is from Columbus, OH, which is within zone B.
- Standard shipping charge: $49.00 per chair
- Actual cost incurred to ship to Columbus: $49.00 per chair
- Zero Phantom or absorb cost.
- Company A will have to absorb a part of the shipping cost in zone C. In this zone, the actual cost of shipping incurred to Company A is higher than the shipping cost charged to the customer.

Scenario I	Scenario II	Scenario III
Standard shipping charge > actual shipping cost	Standard shipping charge = actual shipping cost	Standard shipping charge < actual shipping cost
Phantom cost to customers	Neither party bears additional cost	Freight absorption to sellers

Answer to ParcelExpress

- **Single-Zone Pricing (₹70 flat)**

Revenue = 3,200 × 70 = ₹2,24,000
Total Cost = [(1200×50) + (1000×65) + (700×85) + (300×110)] = ₹2,17,500
Profit = **₹6,500/day**

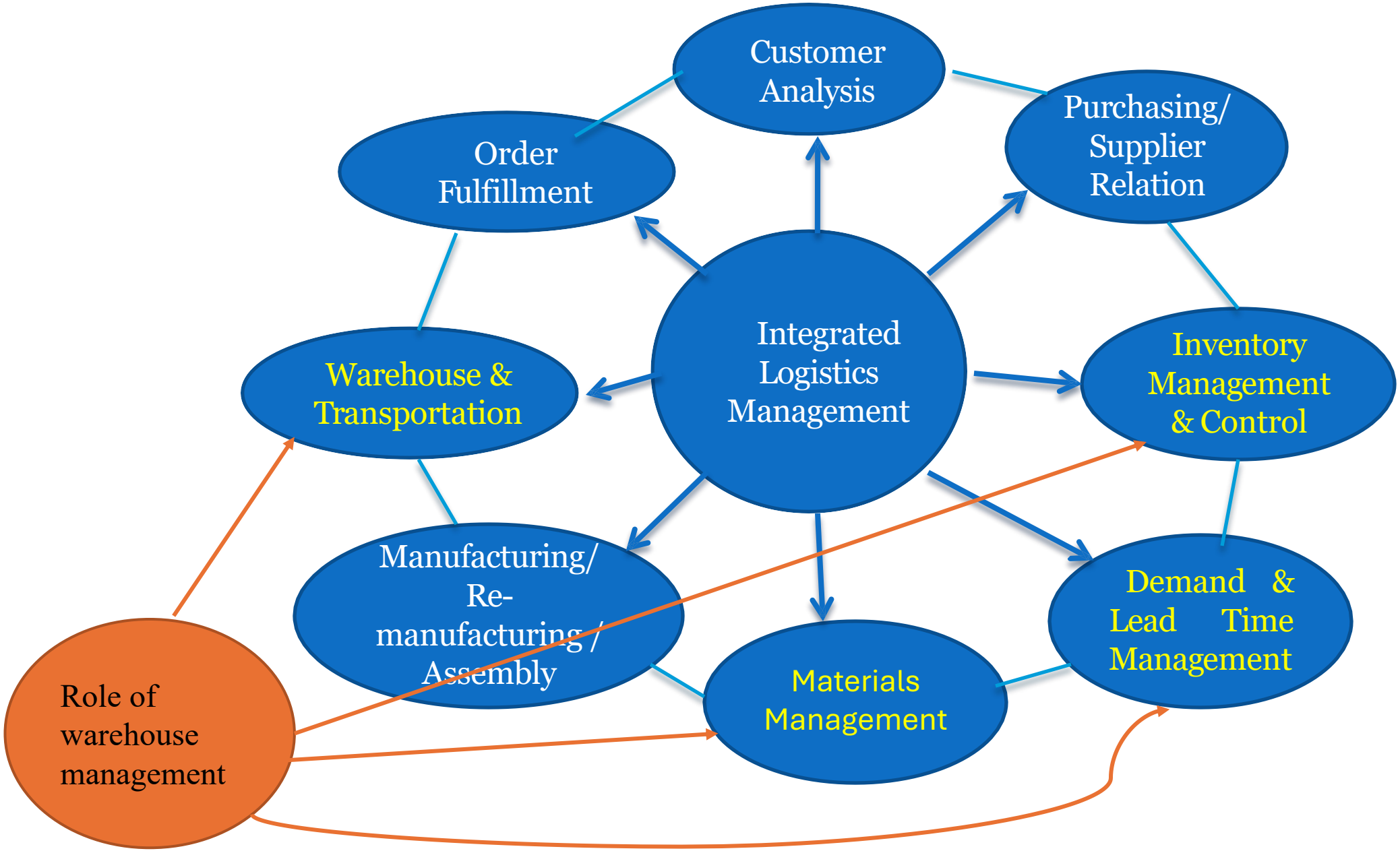
- **Zone Pricing**

zone-wise delivery charges A:55; B:70;C:90;D:120

Revenue = (1200×55) + (1000×70) + (700×90) + (300×120) = ₹2,35,000

Total cost: 2,17,500

Profit: ₹ 17500/day


Customer Analysis
Order Fulfillment
Purchasing/ Supplier Relation
Integrated Logistics Management
Warehouse & Transportation
Inventory Management & Control
Manufacturing/ Re-manufacturing / Assembly
Demand & Lead Time Management
Materials Management
Role of warehouse management

Warehouse

- **Definition** – A designated place where….

- …… goods are stored to balance demand & supply uncertainty
- …….to serve customers in much better way
- …….to consolidate upstream flow and distribute downstream flow as per requirement
- …….to do last mile value addition such as packaging, labeling etc.

- ……..and to reduce last mile distribution cost.

Principle of Warehouse Design (FACT)

- Flow
- Accessibility
- Capacity
- Traceability

Fundamentals of Warehouse Operations

- As discussed previously, warehouse operations will vary by industry, company, strategy, product profile, etc.
- The fundamentals of warehouse operations include:
  - Receiving
  - Putaway
  - Storage
  - Order picking
  - Shipping
- Integrated processes must be designed for each function
- Warehouse Management Systems help to manage processes, tasks and activities, as well as enhance visibility

Fundamentals of Warehouse Operations: Putaway Process Considerations

- Unitization
  - Loose
  - Pallets
  - Slip sheets
  - Totes/bins
- Putaway for storage or immediate shipment?
- Direct Putaway (random location)
- Directed Putaway
- Batched Putaway

Fundamentals of Warehouse Operations: Putaway Process Considerations

- Putaway location decisions
  - Shared pallet position
  - Single product pallet position
  - High volume
  - High value
  - Hazardous Materials
  - Perishables
- Scanned or manual location identification
  - License plate

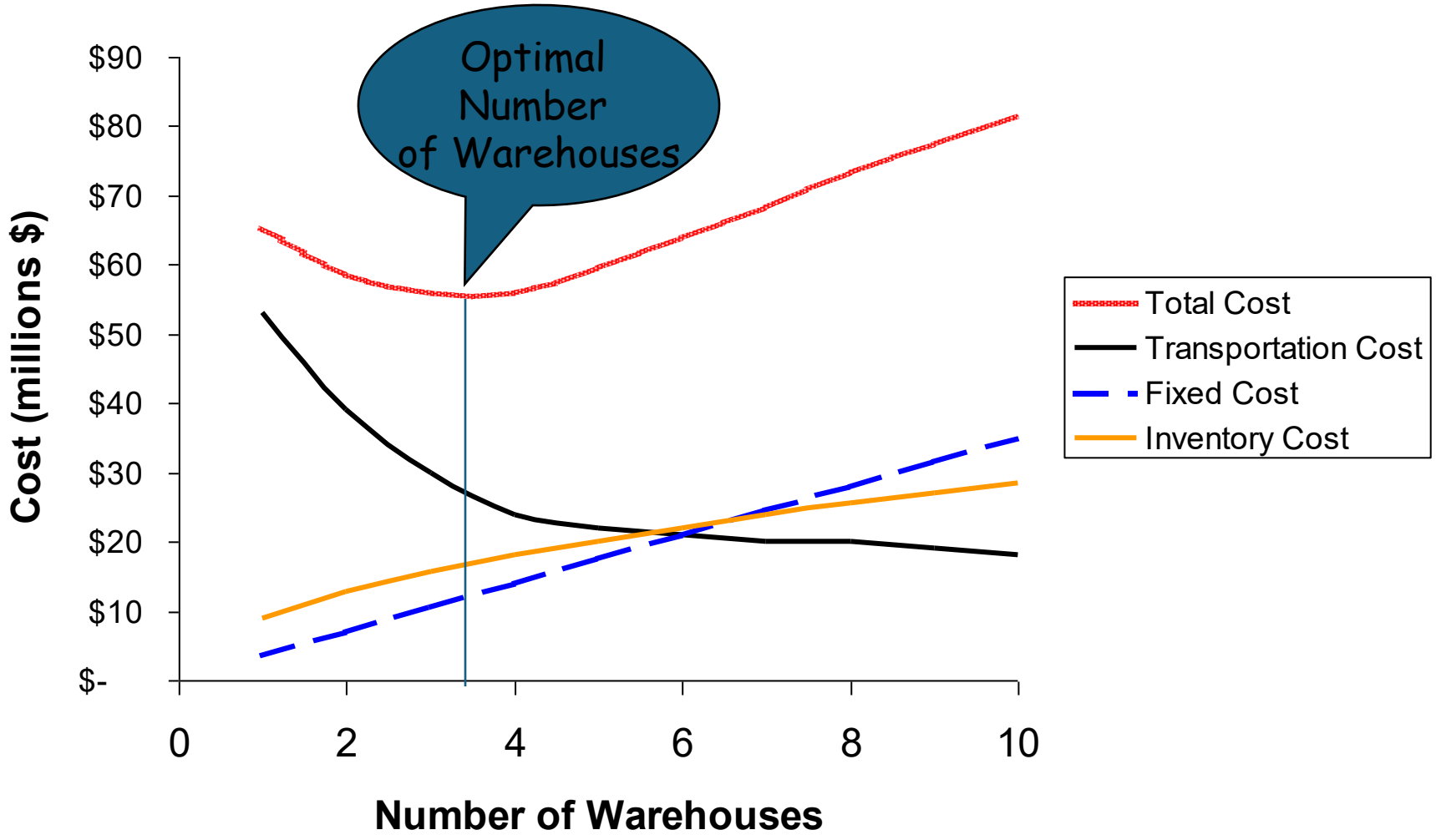
Warehousing Strategy

An integrated warehouse strategy focuses on two questions.

1) The 1st question concerns "how many warehouses should be developed“?.

2) The 2nd question concerns "which type of warehouse should be used to meet market requirements“?.

Number of Warehouses



Warehouse location strategies

1) Market positioned:

a) Order Cycle time
b) Transportation cost
c) Sensitivity of the product
d) Order sizes

2) Product positioned:

a) Perishability of the raw materials
b) Number of products in the product mix
c) Assortments ordered by the customers from the product mix
d) Transportation consolidation rates

3) Intermediately positioned:

Site Selection

- Location of the major markets.
- Nature of the products being distributed.
- Quality & Variety of carriers serving the proposed site.
- Quality & Quantity of labour available.
- Cost of industrial land.

Warehouse location exercise

An e-commerce company wants to locate a warehouse from which it will ship products to four demand points. The location (in the x-y plane) of the four demand points and annual shipment needed by each demand point is given in Table. The company needed an optimized warehouse location to optimize total distance travelled from central warehouse to demand points.

Demand Point	X-coordinate	Y-coordinate	Annual shipment cost
1	5	10	200
2	10	5	150
3	0	12	200
4	12	0	300

Using load distance method of facility.

- The load-distance method is **a mathematical model used to evaluate locations based on proximity factors**.
- The objective is to select a location that minimizes the total weighted loads moving into and out of the facility.
- The distance between two points is expressed by assigning the points to grid coordinates on a map.

Three different way to compute the distance
Actual
Euclidean
Rectilinear

Steps in load distance method

- Find the product of load and the distance from each facililty centers/customer points.
- Find the sum of the product for each perspective locations.
- Select the smallest value of the sum as the most suitable location.

Decision variables

X=x-coordinate of the warehouse

Y=y-coordinate of the warehouse

Objective function

Min load1*sqrt((x1-x)^2+(y1-y)^2)+**load2*sqrt((x2-x)^2+(y2-y)^2)**+load1*sqrt((x3-x)^2+(y3-y)^2)+ load4*sqrt((x4-x)^2+(y4-y)^2)

Company's warehouse to Distributors warehouse transport problem

- Objective: The optimal quantity of cement that should be delivered from company's each warehouse to different distributor's warehouse in order to minimise the transport cost.

Company Warehouse	Distributor Warehouse
Vizag	Raipur
Vijaywada	Bhubaneshwar
Hydrabad	Vizag
	Vijayawada
	Guntur
	Rajahmundry
	Karimnagar

Average shipping cost of per packet

	Vizag	Vijaywada	Hydrabad
Vizag	15	160	100
Vijaywada	160	12	260
Raipur	154	315	56
Bhubnewar	245	410	190
Guntur	130	290	58
Rajahmundri	125	427	204
Karimnagar	215	375	160

- Storage capacity of company warehouse

Vizag>>> 6000

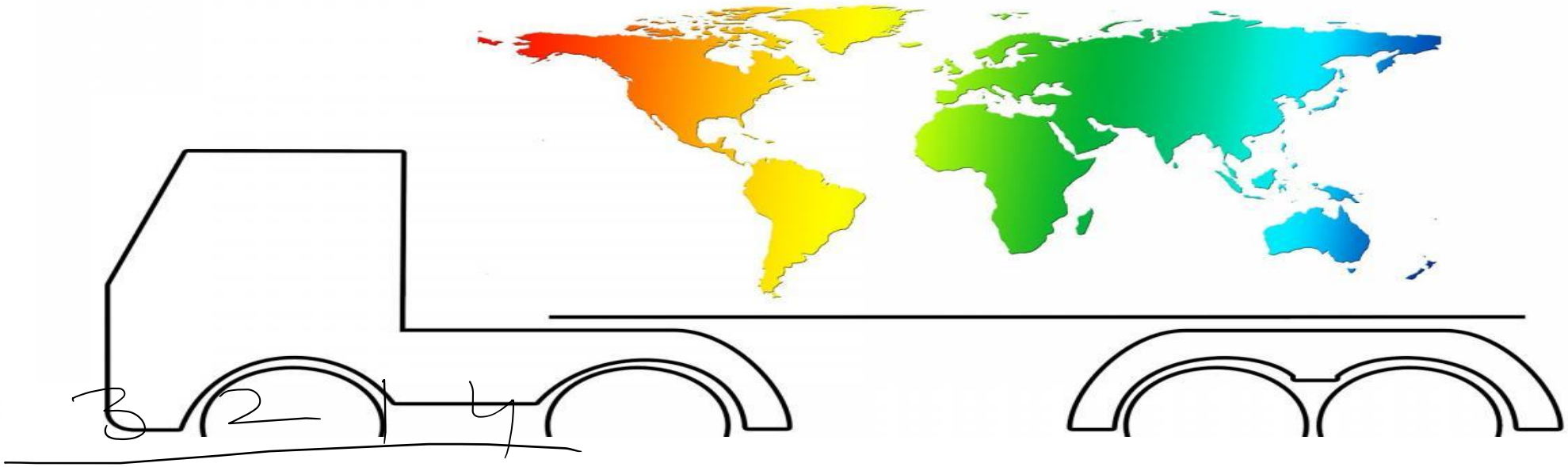
Vijayawada>>> 4000

Hyderabad>>> 5000

Demand at distributors warehouses

- **Vizag 1200**
- Vijayawada 1600
- Raipur 1800
- Bhubaneswar 2100
- Guntur 1300
- Rajahmundry 2900
- Karimnagar 1700

Logistics Outsourcing and Performance



Aalok Kumar

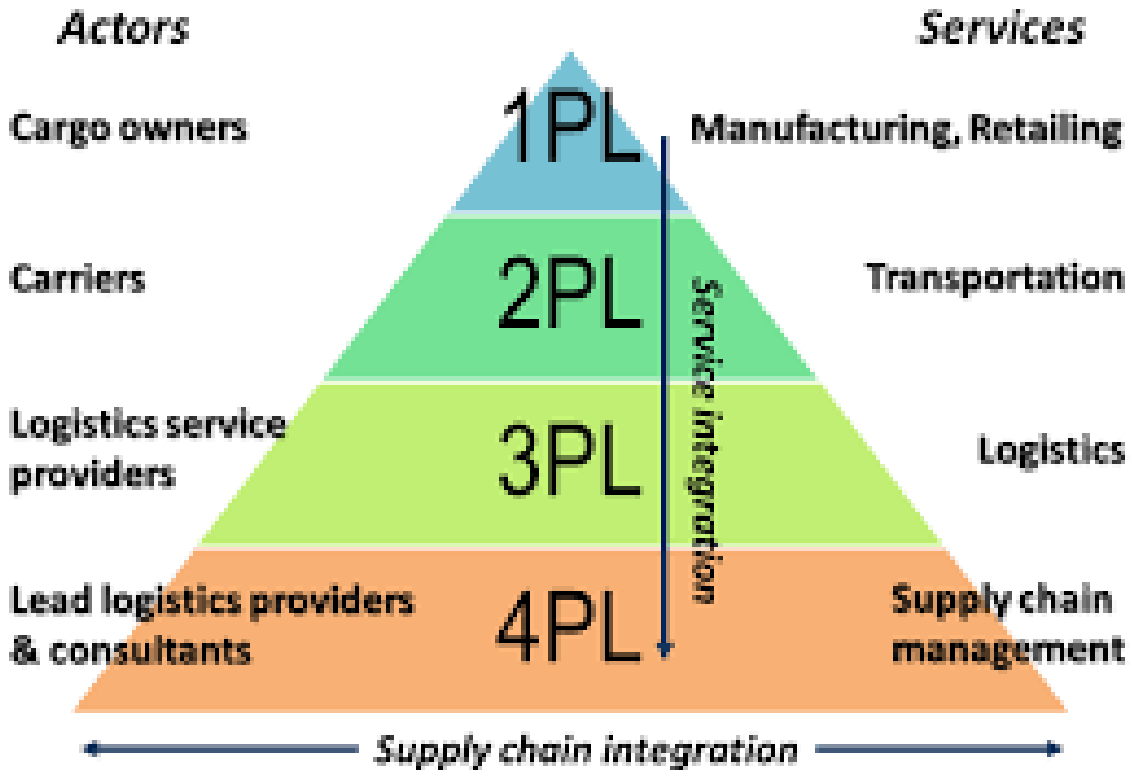
The main issues with outsourcing are

- Balancing operational cost savings with the cost of monitoring suppliers.
- Lower costs due to economies of scale, downsizing or lower labour rates.
- Flexibility to meet changing business and commercial conditions.
- ------
- -------

Outsourcing

- Outsourcing is a make or buy decision which is crucial in determining how a business obtains goods and services.
- Logistics outsourcing is a business practice in which services or job functions are given to a third party, which traditionally were performed in-house by the company's own employees and staff.

- In fact as at 2017, 90% of Fortune 500 companies in the United States alone are working with third-party logistics providers (outsourced), according to a recently released report form Armstrong & Associates.


Actors
Services
Cargo owners
1PL
Manufacturing, Retailing
Carriers
2PL
Transportation
Logistics service providers
3PL
Logistics
Lead logistics providers & consultants
4PL
Supply chain management
Service integration
Supply chain integration


VRL
VRL LOGISTICS LTD


TCI
LEADERS IN LOGISTICS


Mahindra
LOGISTICS


AEGIS


adani
Ports and
Logistics


allcargo


LEAP
...A SOLUTIONS COMPANY


RIVIGO


shadowfax
We Deliver

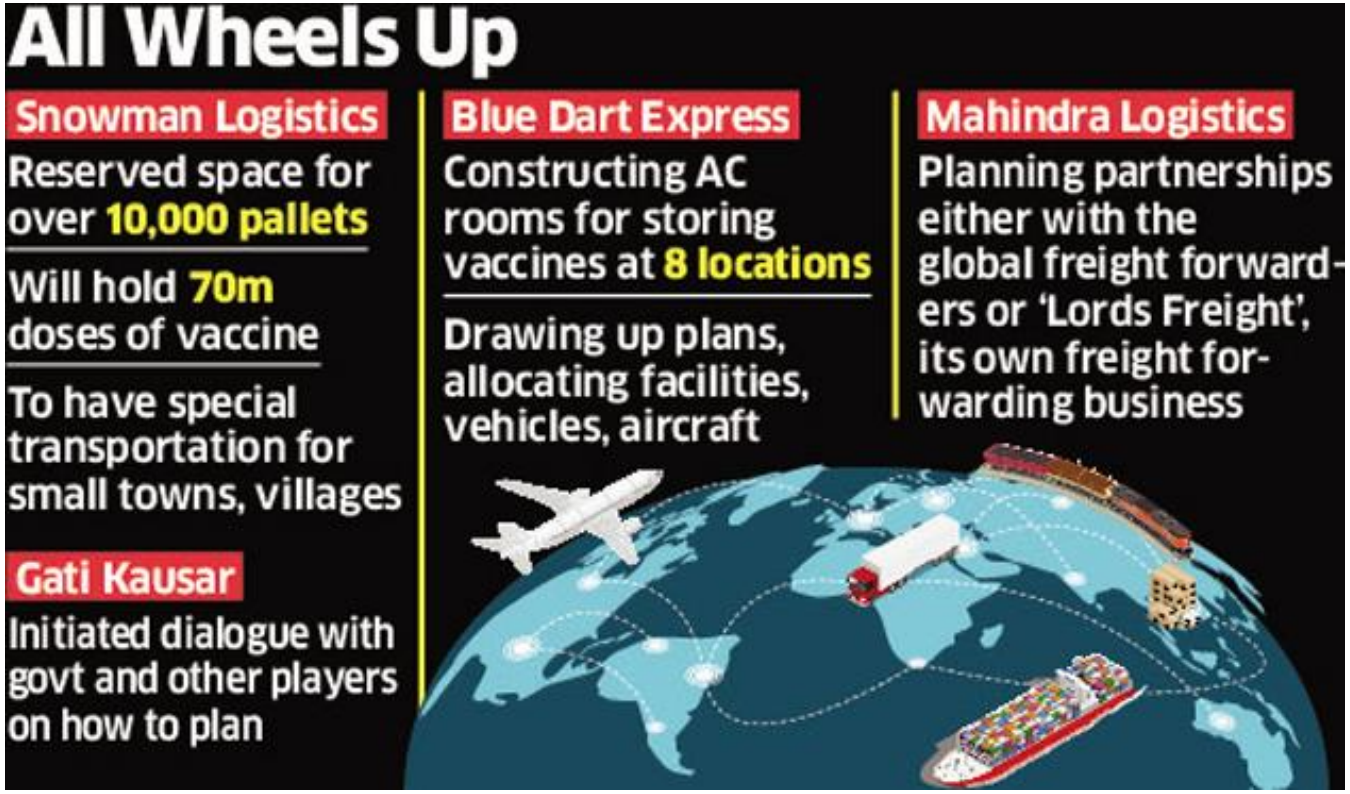

BLACKBUCK


DELHIVERY
Small World


Ecom
Express

Third-party logistics

- Organization's use of third-party businesses to outsource elements of its distribution, warehousing, and fulfillment services.
- 3PL companies targets particular functions within supply management, such as warehousing, transportation, or raw material provision.
- Third-party logistics providers include freight forwarders, courier companies, and other companies integrating & offering subcontracted logistics and transportation services.



Types of 3PL

- Standard 3PL Provider
- Service Developer
- The Customer Adapter
- The Customer Developer

The customer orientation in form of adaptability to changing customer needs, reliability and the flexibility of third-party logistics provider.

Transformation of 3PL

Lead logistics providers

3PL providers without their own assets are called lead logistics providers.

Lead logistics providers have the advantage that they have specialized industry expertise combined with low overhead costs.

They have lower negotiating power and fewer resources than a third-party provider.

Mostly provide consultancy and logistics planning support.

4PL

- Fourth party logistic service providers often have no means of transport and warehouses (non-asset based logistics).
- The fourth party logistic role demands intense involvement from the service supplier in the client's business activities.
- After all, not only does the latter outsource the execution but also the monitoring of the logistic processes.
- Provide end to end solution to logistics activity.



Fifth Party Logistic (5PL)

- 5PLs are business entities those are managing supply chains to supply networks.
- A 5PL guarantees the management of networks of supply chains.
- The industrial actor hires third parties for the supply of strategic, innovative logistical solutions and concepts.
- A fifth party logistic service provider develops and implements, preferably in close consultation with the client, the best possible supply chains or networks.
- Fifth party logistic is often linked to E-business.

3PL vs 4PL

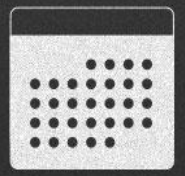
Oftentimes **owns** warehousing or transportation assets



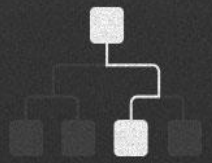
Most often **non-asset** based



Focused on **day-to-day** operations



Focused on **optimizing** the supply chain



One of two points of contact in the supply chain

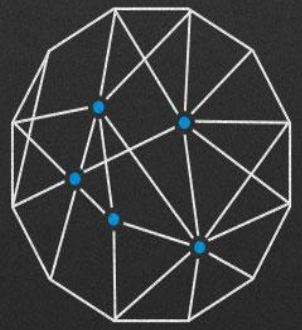


The single point of contact for the full supply chain



A 4PL CAN HELP A RETAILER/ E-COMMERCE:

Allocate and place inventory to **reduce missed opportunities.**



Achieve **same-day delivery** of online purchases regardless of location.



Achieve **same-day inventory** replenishment.



3PL

A fulfillment company packs bananas, stores them & transports them from farm to grocery store



4PL

A logistics company strategically manages a 3PL on behalf of the farmer to package, store & deliver bananas to grocery store



5PL

A logistics company manages a farmers complete supply chain network from production to delivery



1PL

A farmer who delivers eggs to a grocery store for sale.

2PL

A courier delivers eggs from farm to grocery store.

3PL

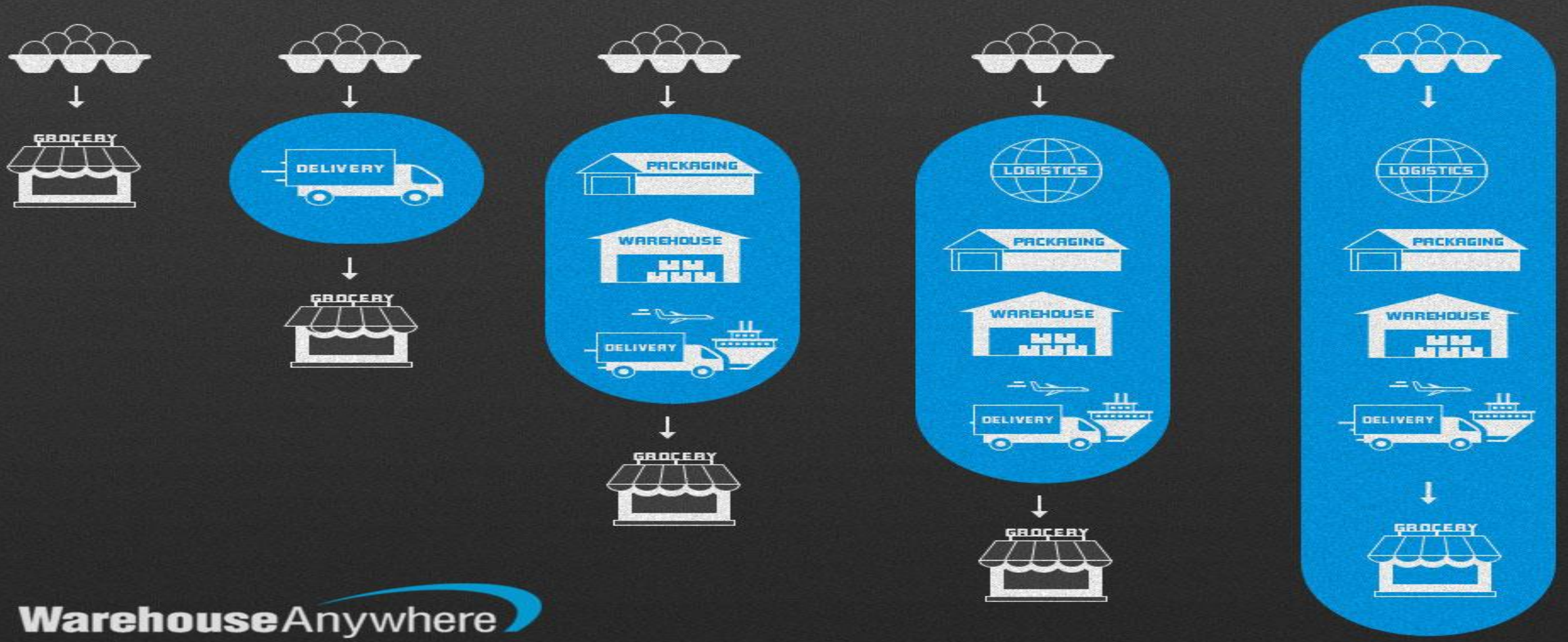
A fulfillment company with a fleet of trucks cartons eggs & transports them from farm to grocery store.

4PL

A logistics company strategically manages a 3PL on behalf of the farmer to package & deliver eggs to grocery store.

5PL

A logistics company manages a farmer's complete supply chain network from production to delivery.



WarehouseAnywhere

LSP selection and assessment model

LSPs selection and assessment model

- In the global competitive scenario, multinational companies (MNCs) not only improve quality of service and increase efficiency, but also decrease costs by means of 3PL.
- However, the assessment and selection of 3PL is a very critical decision, comprising intricacy due to the existence of various imprecisely based criteria.
- Also, uncertainty is an unavoidable part of the information in the decision-making process and its importance in the selection process is relatively high and needs to be carefully considered.

- Due to complexity of FL/RL operations and the lack of available resource, outsourcing **FL/RL** practices to third party reverse logistics provider (3PRLP) is preferred as a strategy.
- The majority of global companies are endorsing and broadly relying on 3PL providers for budget reduction, upgraded strategic alliance and expertise, flexibility in reengineering logistics, access to new technologies and building virtual enterprises focusing on firms' core business.
- But in recent times, increasing supply chain complication, global competition, customer demand for timeliness and flexibility of goods/services delivery have drastically changed the role of 3PLs.

MCDA methodology

MCDA is an umbrella term for a range of tools and methodologies. The level of complexity, interaction with the decision maker and level of detail utilised in the decision-making process can vary substantially.

In general the decision maker follows the same process:

1. Identify multiple criteria on which to base their decision;
2. Identify multiple alternative solutions to their decision;
3. Provide (subjective) ranking or weighting of criteria; and
4. Provide values, rankings or weighting of alternatives for each criteria.

MCDA methodology

The Analytical Hierarchy Process (AHP)

- AHP was developed in the late 1970s. Today it is the most widely used MCDA method.

- AHP generates all criteria weighting and alternative preference within each criteria by eliciting these values from the decision maker through a series of pairwise comparisons, as opposed to utilising numerical values directly.

- Thus, a complex decision is reduced to a series of simpler ones, between pairs of alternative values within criteria or between pairs of criteria.
- Hence, it becomes important to generate an optimised hierarchy of criteria and alternatives, to reduce the number of pairwise decisions.

MCDA methodology

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AHP

Step 1: Construct the problem hierarchy

Model, usually visually, the problem decision identifying relationships between criteria and alternatives.

Step 2: Pairwise comparison of criteria

Undertake pairwise comparison between criteria, identifying decision maker preference for criteria on which alternatives are evaluated.

Step 3: Pairwise comparison of alternatives within each criterion

Undertake pairwise comparison between alternatives based on their performance within each criterion.

Step 4: Compute the vector of criteria weights

From a matrix of pairwise comparison results AHP utilises a variety of matrix transformations to calculate criteria weight vectors representing normalised criteria weightings.

AHP Example

Pairwise Comparisons

Within AHP pairwise comparison is the process of comparing entities in pairs so as to judge which is preferred and by how much. Comparisons are undertaken to determine criteria weighting and also assess the value or score of different alternatives within each criteria.
A 9-point scale to elicit the scale of preference from a decision maker:
the more preferable entity within the pair scores

1 when it is showing no preference
3 when it is showing moderate preference
5 when it is showing strong preference
7 when it is showing very strong preference
9 when it is showing extreme preference.

The less preferable entity within the pair scores the inverse, for example the less preferable entity where the more preferable entity shows very strong preference would score 1/7.

AHP

	Criteria A1	Criteria A2	Criteria A3	Criteria A4
Criteria A1	1	5	3	7
Criteria A2	1/5	1	1/3	3
Criteria A3	1/3	3	1	7
Criteria A4	1/7	1/3	1/7	1

The results of the matrix would provide the normalised criteria weights for criteria A1 to A4. Similar matrices would be completed for criteria B1 to B4, for C1 to C4 and also one comparing criteria A, B and C. Finally, pairwise comparisons would be undertaken to fill matrices for each criteria comparing the performance of each alternative within that criteria.

AHP

Consistency across pairwise comparisons

The consistency of the decision maker across a number of pairwise comparisons is a significant complexity. Consider the very simple comparison of three criteria: A, B and C.

If the decision maker judges A to be more preferable than B, and

A to be less preferable than C then the decision maker must not judge **B to be more preferable than C.**

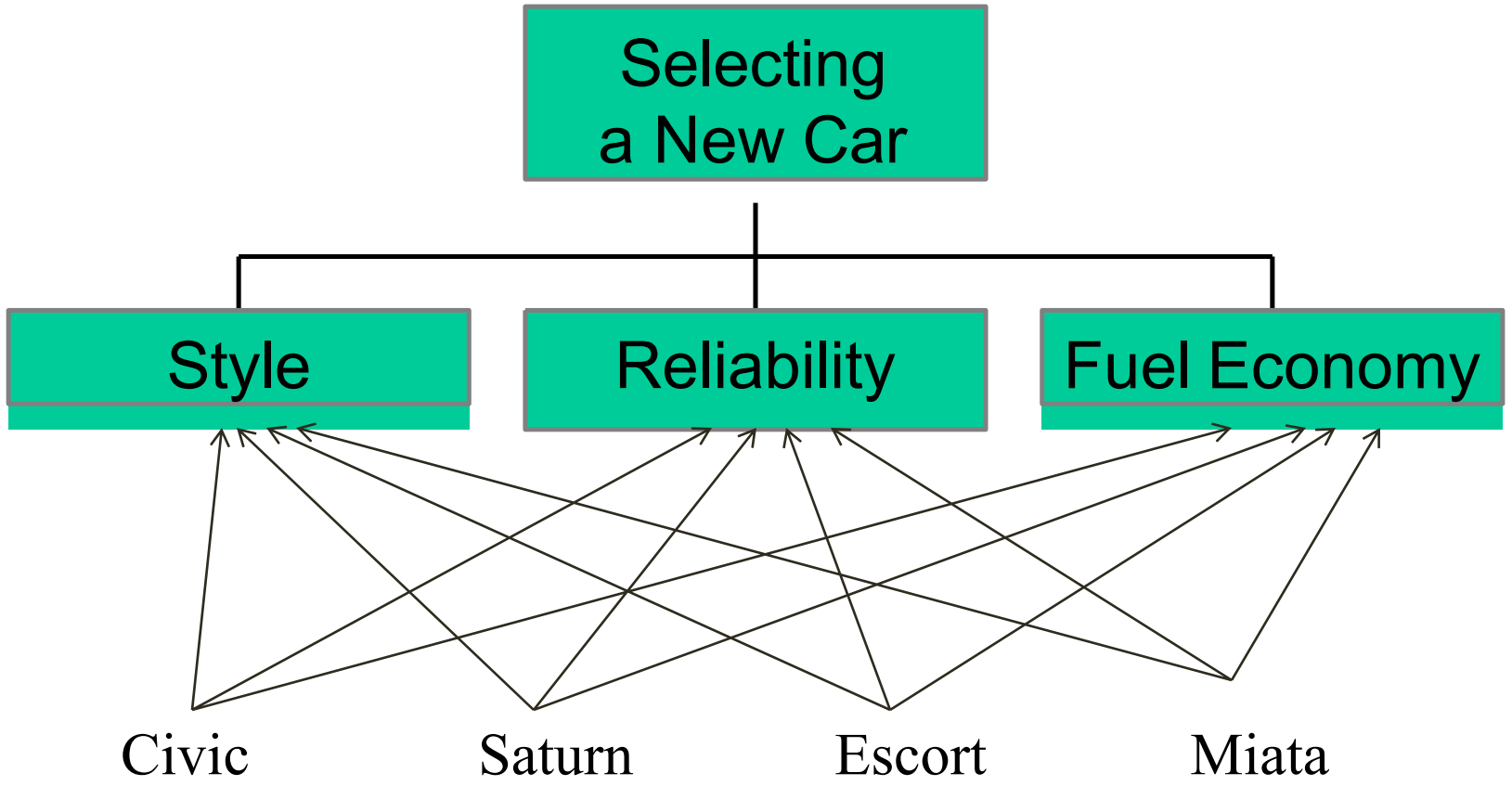
In a group that contains a large number of pairwise comparisons or where the difference is between moderate and very strong preference it can be seen that lack of consistency is a largely inevitable consequence of complex decision processes within AHP.

The AHP method attempts to address the issue of consistency by implementing a consistency index that is a function of opposing comparisons.

Above a threshold a lack of consistency is highlighted and no analysis results are presented.

An unfortunate consequence is that decision makers begin to fulfil pairwise comparisons not on their actual judgements but rather in order to maintain acceptable consistency.

AHP Problem



Alternative courses of action

Ranking Scale for Criteria and Alternatives

Intensity of importance	Definition	Explanation
1	Equal importance	Two factors contribute equally to the objective
3	Somewhat more important	Experience and judgement slightly favour one over the other.
5	Much more important	Experience and judgement strongly favour one over the other.
7	Very much more important	Experience and judgement very strongly favour one over the other. Its importance is demonstrated in practice.
9	Absolutely more important.	The evidence favouring one over the other is of the highest possible validity.
2,4,6,8	Intermediate values	When compromise is needed

Ranking of criteria

	Style	Reliability	Fuel Economy
Style	1	1/2	3
Reliability	2	1	4
Fuel Economy	1/3	1/4	1

Ranking of priorities

- Consider [Ax = λ_{max}x] where
  - A is the comparison matrix of size n n, for n criteria, also called the priority matrix.
  - x is the Eigenvector of size n 1, also called the priority vector.
  - λ_{max} is the Eigenvalue.

- To find the ranking of priorities, namely the Eigen Vector X:

  1) Normalize the column entries by dividing each entry by the sum of the column.

  2) Take the overall row averages.

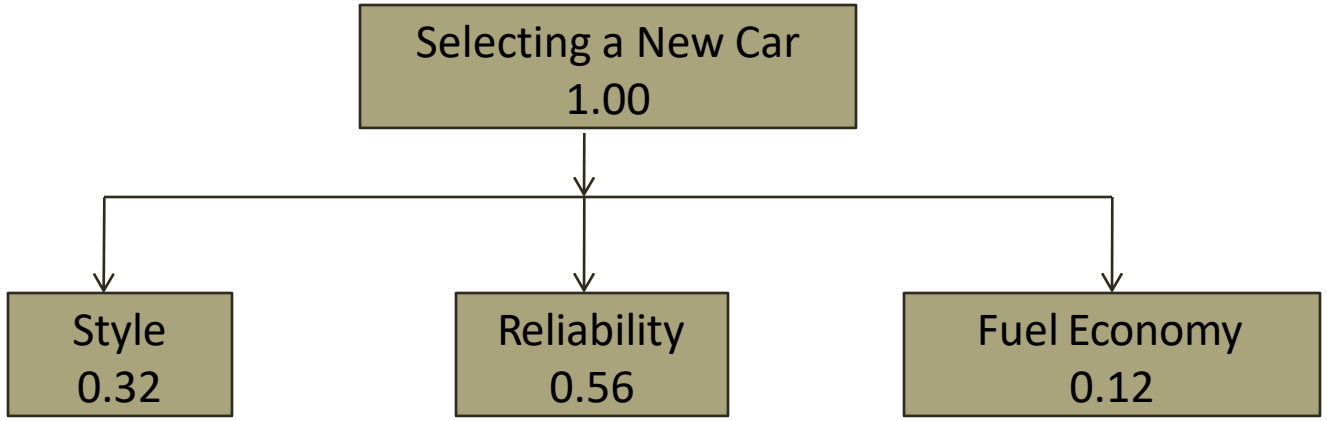
$$A = \begin{bmatrix} 1 & 0.5 & 3 \\ 2 & 1 & 4 \\ 0.33 & 0.25 & 1.0 \end{bmatrix} \xrightarrow{\text{Normalized Column Sums}} \begin{bmatrix} 0.30 & 0.28 & 0.37 \\ 0.60 & 0.57 & 0.51 \\ 0.10 & 0.15 & 0.12 \end{bmatrix} \xrightarrow{\text{Row averages}} X = \begin{bmatrix} 0.32 \\ 0.56 \\ 0.12 \end{bmatrix}$$

Priority vector

Column sums 3.33 1.75 8.00 1.00 1.00 1.00

Criteria weights

- Style .32
- Reliability .56
- Fuel Economy .12



Calculation of Consistency Ratio

- The next stage is to calculate λ_{max} to lead to the Consistency Index and the Consistency Ratio.
- Consider [$Ax = \lambda_{max} x$] where x is the Eigenvector.

$$\underset{A}{\begin{bmatrix} 1 & 0.5 & 3 \\ 2 & 1 & 4 \\ 0.333 & 0.25 & 1.0 \end{bmatrix}} \underset{X}{\begin{bmatrix} 0.32 \\ 0.56 \\ 0.12 \end{bmatrix}} = \underset{Ax}{\begin{bmatrix} 0.96 \\ 1.68 \\ 0.36 \end{bmatrix}} = \lambda_{max} \underset{x}{\begin{bmatrix} 0.32 \\ 0.56 \\ 0.12 \end{bmatrix}}$$

$\lambda max = average\{0.96/0.32, 1.68/0.56, 0.36/0.12\} = 3.00$

Consistency index , CI is found by

$$CI = (\lambda max - n)/(n-1) = (3.0-3)/(3-1) = 0.00$$

$R(\lambda) = ||Ax|| / ||x||$

The Rayleigh quotient is used in the min-max theorem to get exact values of all eigenvalues.

C.R. = C.I./R.I. where R.I. is the random index

Matrix size	Random consistency index (RI)
1	0.00
2	0.00
3	0.58
4	0.90
5	1.12
6	1.24
7	1.32
8	1.41
9	1.45
10	1.49

C.I. = 0 n = 3

R.I. = 0.58(from table)

So, C.R. = C.I./R.I. = 0/0.58 = 0.04

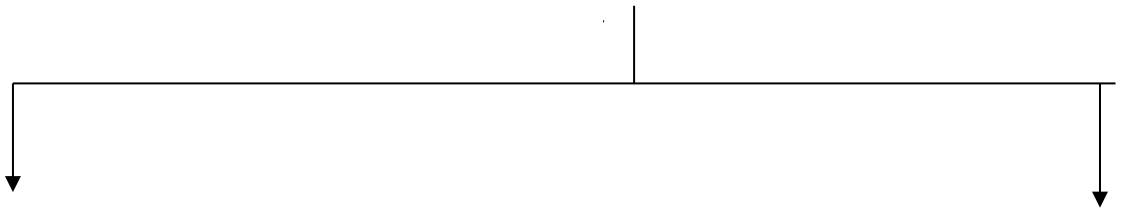
C.R. $\leq$ 0.1 indicates sufficient consistency for decision.

Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

Basic Concept

- **The chosen Alternative should have the shortest distance from the ideal solution and the farthest from the negative-ideal solution (anti-ideal point).**

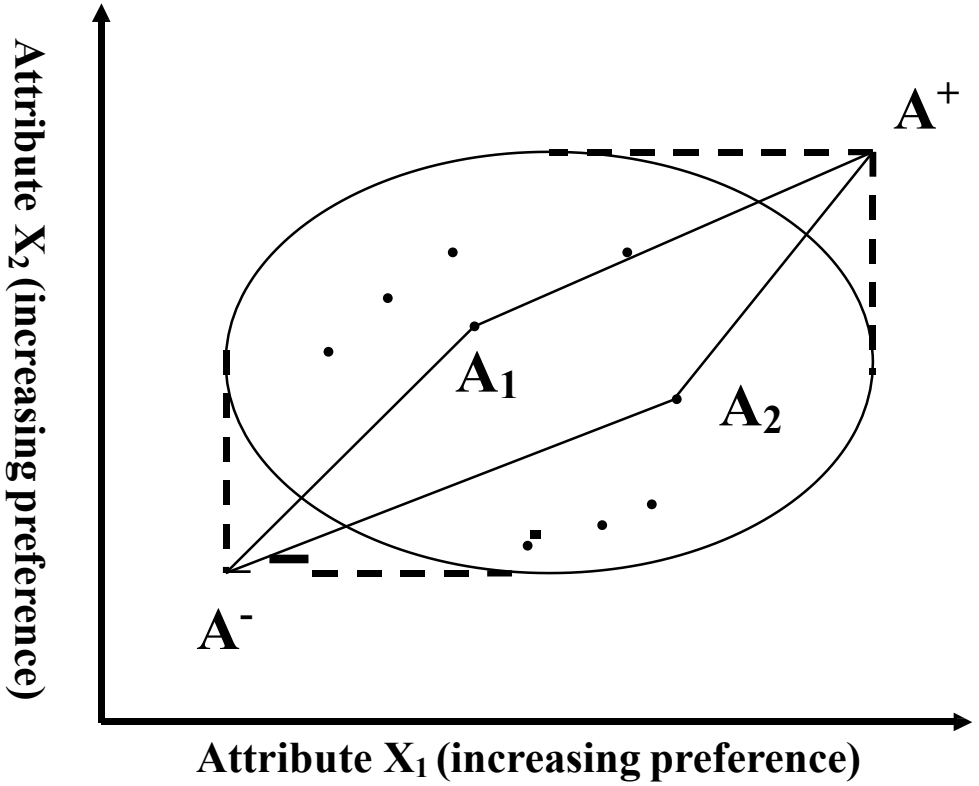
CRITERIA



TANGIBLE
(measurable in Rs., percentage, year, Rs/year, …)

INTANGIBLE
(not measurable on a well defined scale)

Basic Concept …



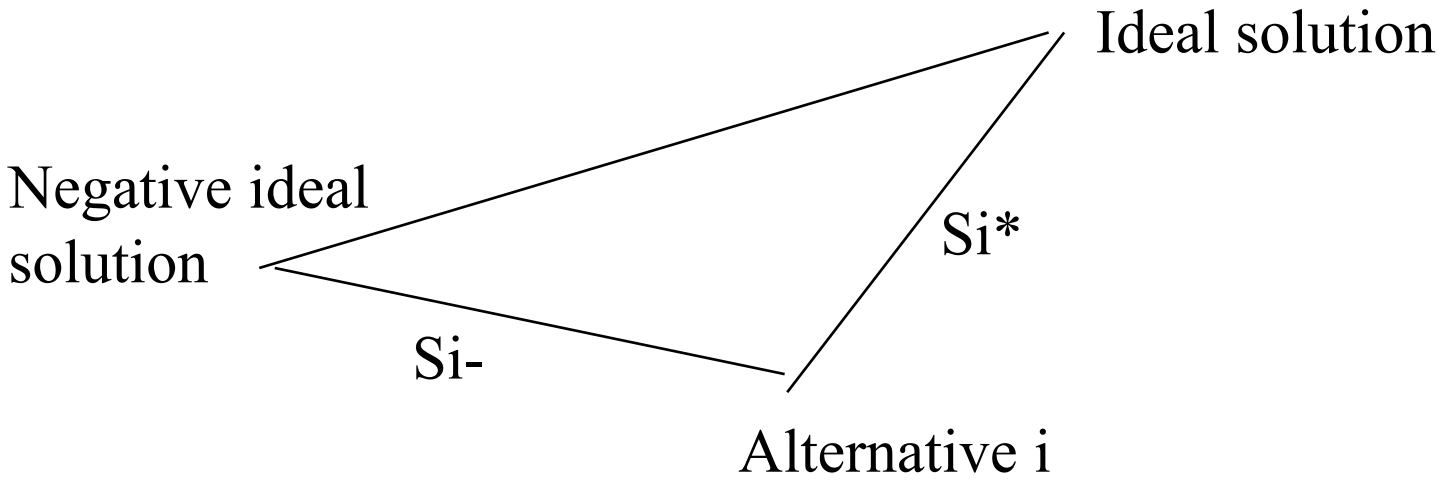
It is very difficult to justify the selection of A_1 or A_2

RELATIVE CLOSENESS TO IDEAL SOLUTION

Step 6 For each alternative determine the relative closeness to the ideal solution

(C_i^*, i=1,…,m) as

$$C_i^* = S_i^- / (S_i^* + S_i^-)$$



Decision Matrix

- **m Alternative, n Attributes (or criteria)**

$$D = \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \\ \cdot \\ \cdot \\ \cdot \\ A_m \end{array} \begin{array}{c} \begin{array}{cccccc} x_1 & x_2 & x_3 & \cdot & \cdot & \cdot & x_n \end{array} \\ \begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdot & \cdot & \cdot & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdot & \cdot & \cdot & x_{2n} \\ x_{31} & x_{32} & x_{33} & \cdot & \cdot & \cdot & x_{3n} \\ \cdot & \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & \cdot & & & & \cdot \\ x_{m1} & x_{m2} & x_{m3} & \cdot & \cdot & \cdot & x_{mn} \end{bmatrix} \end{array}$$

Important points

- **Each Attribute in the Decision Matrix takes either monotonically increasing or monotonically decreasing utility.**
- **A Set of Weights for the Attributes is required.**
- **Any Outcome which is expressed in a non-numerical way, should be quantified through the appropriate scaling technique**

ComputationalSteps-1

- **Construct the Normalized Decision Matrix**
  - **To transform the various attribute dimensions into non-dimensional attributes, which allows comparison across the attributes**

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$$

Steps-2

- **Construct the Weighted Normalized Decision Matrix**

$$V = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1j} & \cdots & v_{1n} \\ \vdots & \vdots & & \vdots & & \vdots \\ v_{i1} & v_{i2} & & v_{ij} & & v_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ v_{m1} & v_{m2} & \cdots & v_{mj} & \cdots & v_{mn} \end{bmatrix} = \begin{bmatrix} w_1 r_{11} & w_2 r_{12} & \cdots & w_j r_{ij} & \cdots & w_n r_{1n} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_1 r_{i1} & w_2 r_{i2} & & w_j r_{ij} & & w_n r_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_1 r_{m1} & w_2 r_{m2} & \cdots & w_j r_{mj} & \cdots & w_n r_{mn} \end{bmatrix}$$

Steps-3

- **Determine Ideal and Negative-Ideal Solutions**

$$A^{+} = \{(\max_i v_{ij} \,|\, j \in J), (\min_i v_{ij} \,|\, j \in J') \,|\, i = 1,2,...m\}$$
$$= \{v_1^{+}, v_2^{+}, ..., v_j^{+}, ..., v_n^{+}\}$$

$$A^{-} = \{(\min_i v_{ij} \,|\, j \in J), (\max_i v_{ij} \,|\, j \in J') \,|\, i = 1,2,...m\}$$
$$= \{v_1^{-}, v_2^{-}, ..., v_j^{-}, ..., v_n^{-}\}$$

$$where \quad J = \{j = 1,2,...,n \,|\, j \quad associated \quad with \quad benefit \quad criteria\}$$
$$J' = \{j = 1,2,...,n \,|\, j \quad associated \quad with \quad \cos t \quad criteria\}$$

Steps-4

- **Calculate the Separation Measure:**
  - **Ideal Separation**

$$s_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2} \qquad i = 1,2,...,m$$

  - **Negative-Ideal Separation**

$$s_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2} \qquad i = 1,2,...,m$$

Steps-5

- **Calculate the Relative Closeness to the Ideal Solution**

$$c_i^* = \frac{s_i^-}{(s_i^+ + s_i^-)}, \quad 0 < c_i^+ < 1, \quad i = 1,2,...,m$$

$$c_i^* = 1 \quad if \quad A_i = A^+$$

$$c_i^* = 0 \quad if \quad A_i = A^-$$

Steps-6

- **Rank the preference order**
  - **A set of alternatives can now be preference ranked according to the descending order of C_i^***

An example of using TOPSIS Method

Weights are computed with the AHP method

	Style	Reliability	Fuel Eco.	Cost
Weight	0.1	0.4	0.3	0.2
Civic	7	9	9	8
Saturn	8	7	8	7
Ford	9	6	8	9
Mazda	6	7	8	6

Steps of TOPSIS

- Step 1: Calculate $(\Sigma x^2_{ij})^{1/2}$ for each column and divide each column by that to get r_{ij}

	Style	Rel.	Fuel	Cost
Civic	0.46	0.61	0.54	0.53
Saturn	0.53	0.48	0.48	0.46
Ford	0.59	0.41	0.48	0.59
Mazda	0.40	0.48	0.48	0.40

Steps ofTOPSIS

- Step 2 : Multiply each Column by w_j to get V_{ij}.

	Style	Rel.	Fuel	Cost
Civic	0.046	0.244	0.162	0.106
Saturn	0.053	0.192	0.144	0.092
Ford	0.059	0.164	0.144	0.118
Mazda	0.040	0.192	0.144	0.080

For Cost type criterion look for minimum and for benefit type criterion look for maximum criterion

Steps of TOPSIS

Step 3 (a): Determine Ideal Solution A*.

A* = {0.059, 0.244, 0.162, 0.080}

	Style	Rel.	Fuel	Cost
Civic	0.046	0.244	0.162	0.106
Saturn	0.053	0.192	0.144	0.092
Ford	0.059	0.164	0.144	0.118
Mazda	0.040	0.192	0.144	0.080

Steps of TOPSIS

Step 3 (b): Find Negative Ideal Solution A^-.

$A^- = \{0.040, 0.164, 0.144, 0.118\}$

	Style	Rel.	Fuel	Cost
Civic	**0.046**	**0.244**	**0.162**	**0.106**
Saturn	**0.053**	**0.192**	0.144	**0.092**
Ford	**0.059**	0.164	0.144	0.118
Mazda	0.040	**0.192**	0.144	**0.080**

↓ ↑

Steps of TOPSIS

Step 4 (a): Determine Separation From Ideal Solution

$A^* = \{0.059, 0.244, 0.162, 0.080\}$

$S_i^* = [\ \Sigma\ (v_j^* - v_{ij})^2\ ]^{1/2}$ for each row j

	Style	Rel.	Fuel	Cost
Civic	$(.046-.059)^2$	$(.244-.244)^2$	$(0)^2$	$(.026)^2$
Saturn	$(.053-.059)^2$	$(.192-.244)^2$	$(-.018)^2$	$(.012)^2$
Ford	$(.053-.059)^2$	$(.164-.244)^2$	$(-.018)^2$	$(.038)^2$
Mazda	$(.053-.059)^2$	$(.192-.244)^2$	$(-.018)^2$	$(.0)^2$

Steps of TOPSIS

Step 4 (a): Determine Separation From Ideal Solution S_i^*

	$\Sigma(v_j^* - v_{ij})^2$	$S_i^* = [\Sigma (v_j^* - v_{ij})^2]^{1/2}$
Civic	**0.000845**	0.029
Saturn	**0.003208**	0.057
Ford	**0.008186**	0.090
Mazda	**0.003389**	0.058

Steps of TOPSIS

Step 4: Determine Separation From Negative Ideal Solution S_i^-

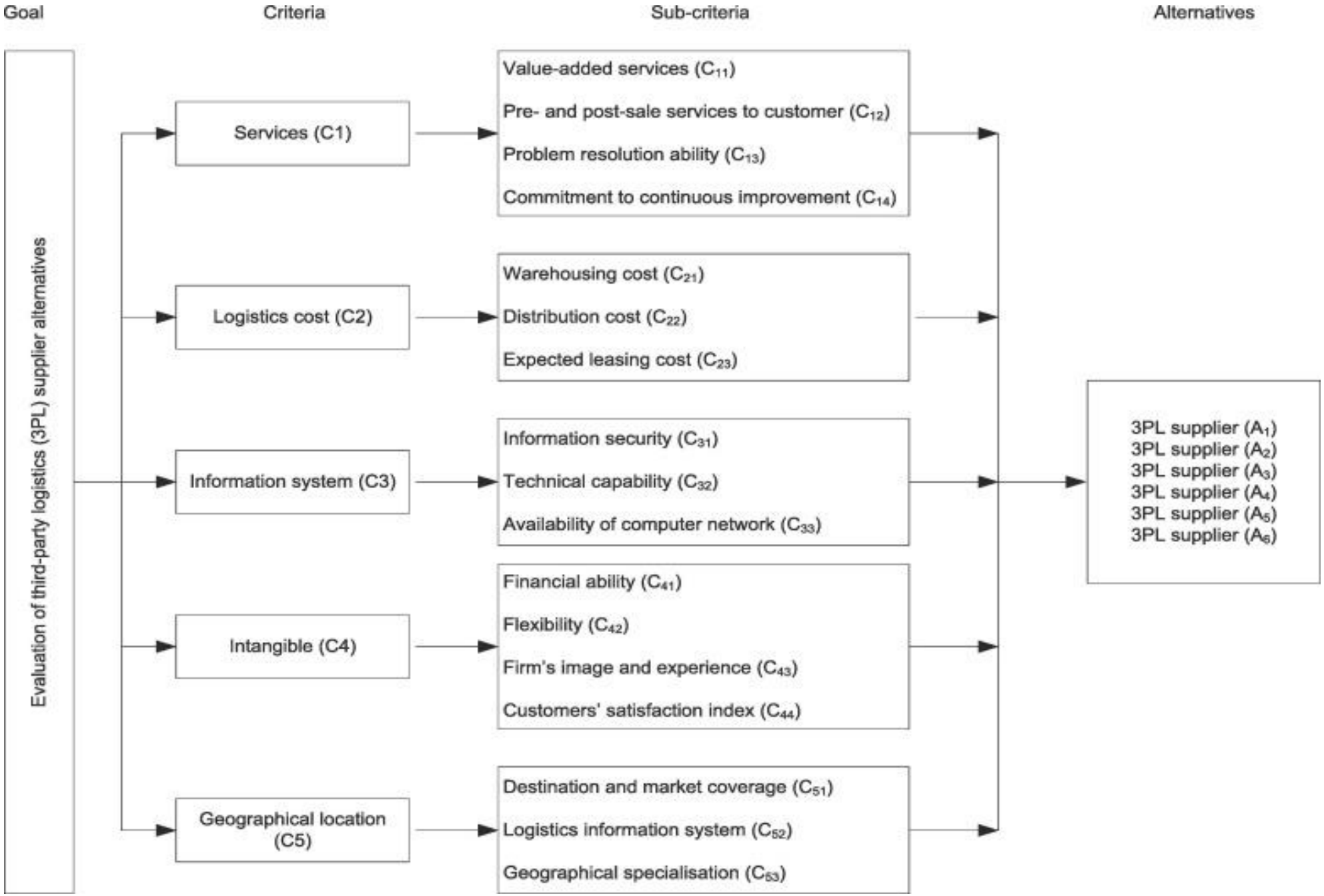
	$\Sigma(v_j^- - v_{ij})^2$	$Si^- = [\Sigma (v_j^- - v_{ij})2]^{½}$
Civic	**0.006904**	0.083
Saturn	**0.001629**	0.040
Ford	**0.000361**	0.019
Mazda	**0.003389**	0.058

Steps of TOPSIS

Step 5: Calculate the relative closeness to the ideal solution $Ci^* = S^-_i / (S^*_i + S^-_i)$

	$S^-_i/(S^*_i + S^-_i)$	Ci^*		
Civic	**0.083/0.112**	0.74	←	BEST
Saturn	**0.040/0.097**	0.41		
Ford	**0.019/0.109**	0.17	←	WORSE
Mazda	**0.047/0.105**	0.45		

3PLSP selection problem



Source: https://www.sciencedirect.com/science/article/pii/S0360835218304935

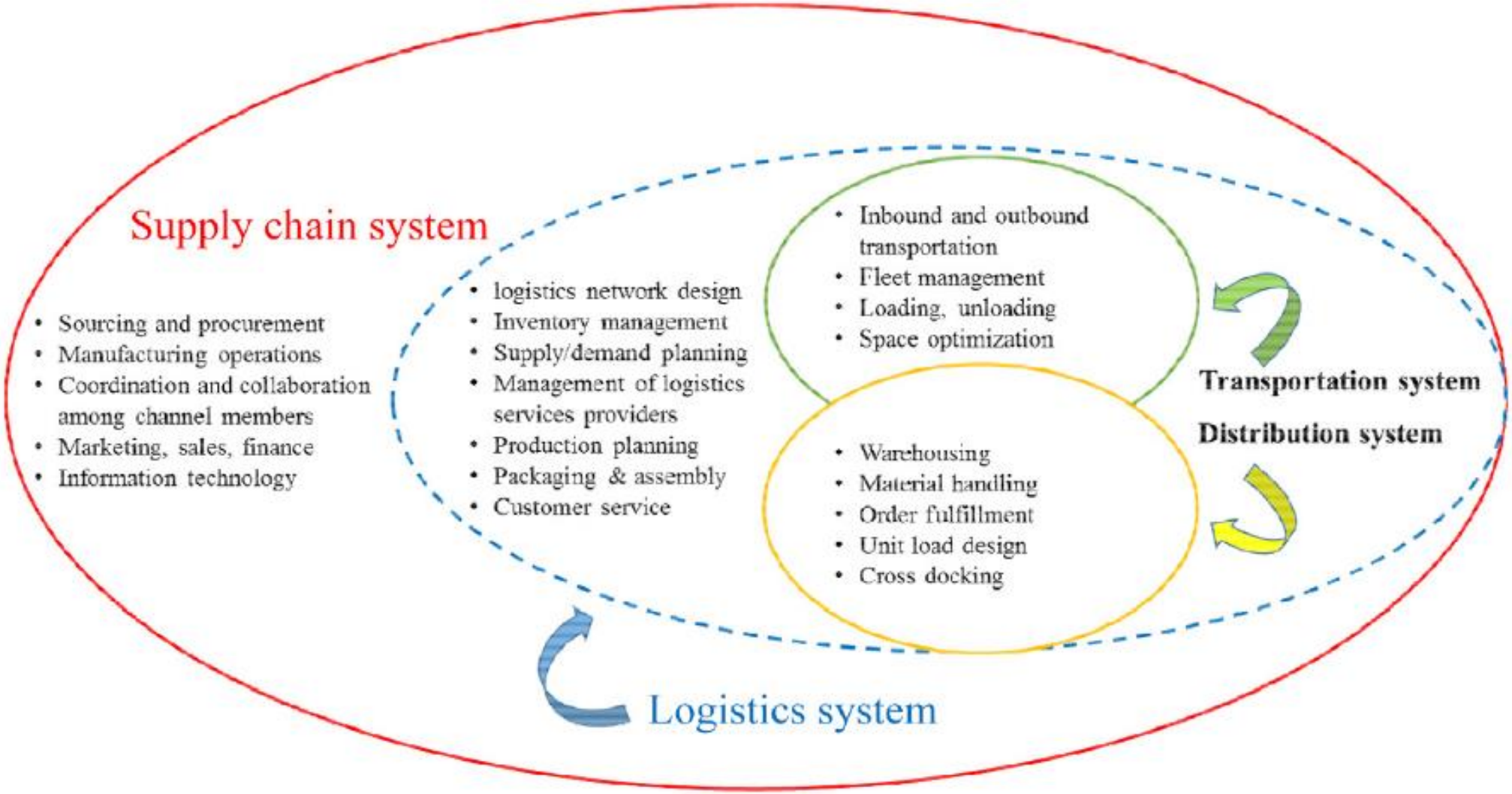
Transportation Management



Discussion points

- Transportation role in logistics
- Transportation operational planning
- Transportation Sourcing
- Transportation routing
- ICD and continuation
- International transport
- Terminal operations management

What is the transportation in logistics and its importance



Transportation mode

- There are five major modes of transportation—air, rail, road, maritime, and pipeline.
- Each of these modes of transportation must have a number of employees directly working on transportation operations and additional employees of service companies that provide other services directly related to these same operations.
- These logistics service personnel play a pivotal role in ensuring that fleet operations are efficient.

Objective of transport analyst

- First, logisticians, professionals in the field of logistics management, act as subject matter experts on transportation system design and components within a given supply chain.
- Second, the logistician must act as a catalyst for improvement in the operating systems and decision making associated with supply chain management.
- .....

Transport planning for logistics operations

- What carriers should I partner with and how?
- How will seasonality affect my carrier assignments?
- Should I use dedicated or private fleets?
- Which carriers provided quality services in the past?
- Should I use pool points, cross-docks, or multistop routes?
- How can I quickly secure rates for a new DC/plant/lane?
- …….

- What lanes are having performance problems?
- Which carriers are complying to or exceeding their contracts?
- Where should I establish a seasonal contract?
- Which carrier should I tender this load to?
- …….

Transportation process

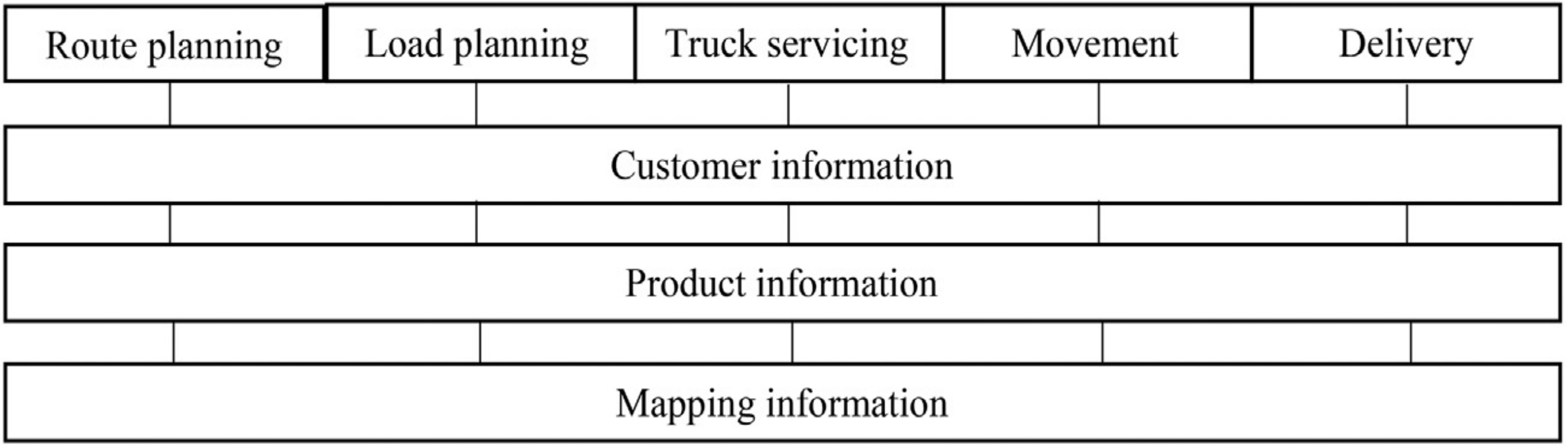
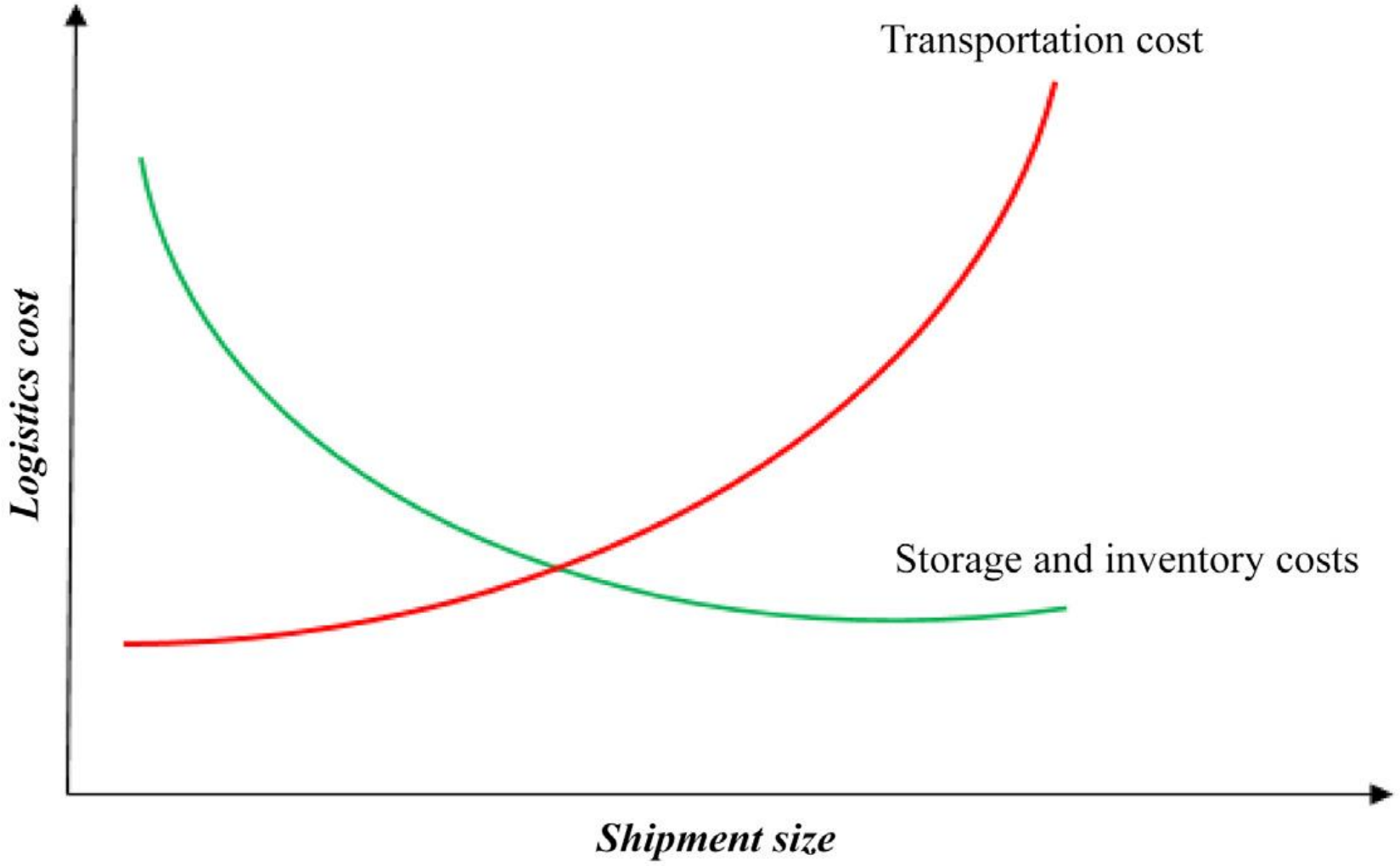


Figure: Component of transportation process

Transportation cost analysis (TCA)



Other important factors for transport analysis

Following are some of the factors to consider when transporting goods:

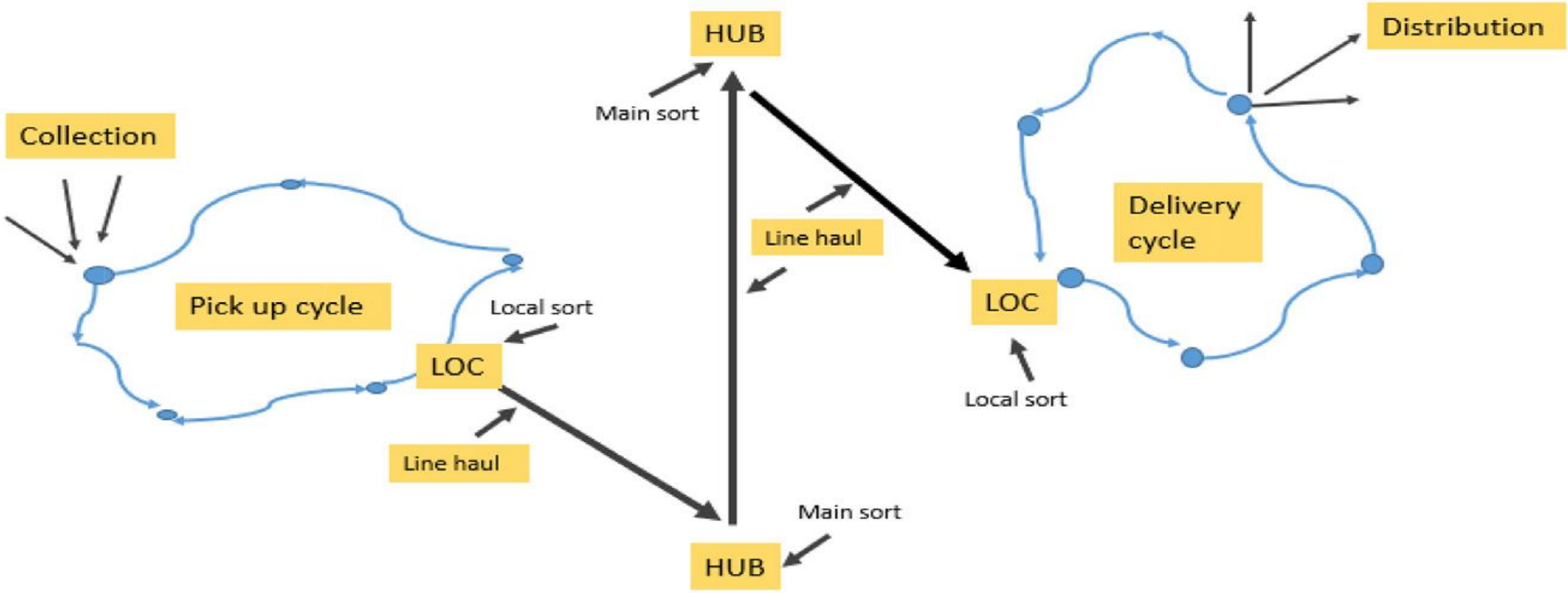
- Type of goods (perishable, nonperishable, fragile, irregular size, dangerous).
- Availability of goods and time needed to source them
- Existing transportation infrastructures and equipment
- Access to suppliers and logistics providers
- Time needed to deliver goods to the client
- Customs requirements
- Volume of shipment
- Modes of transportation

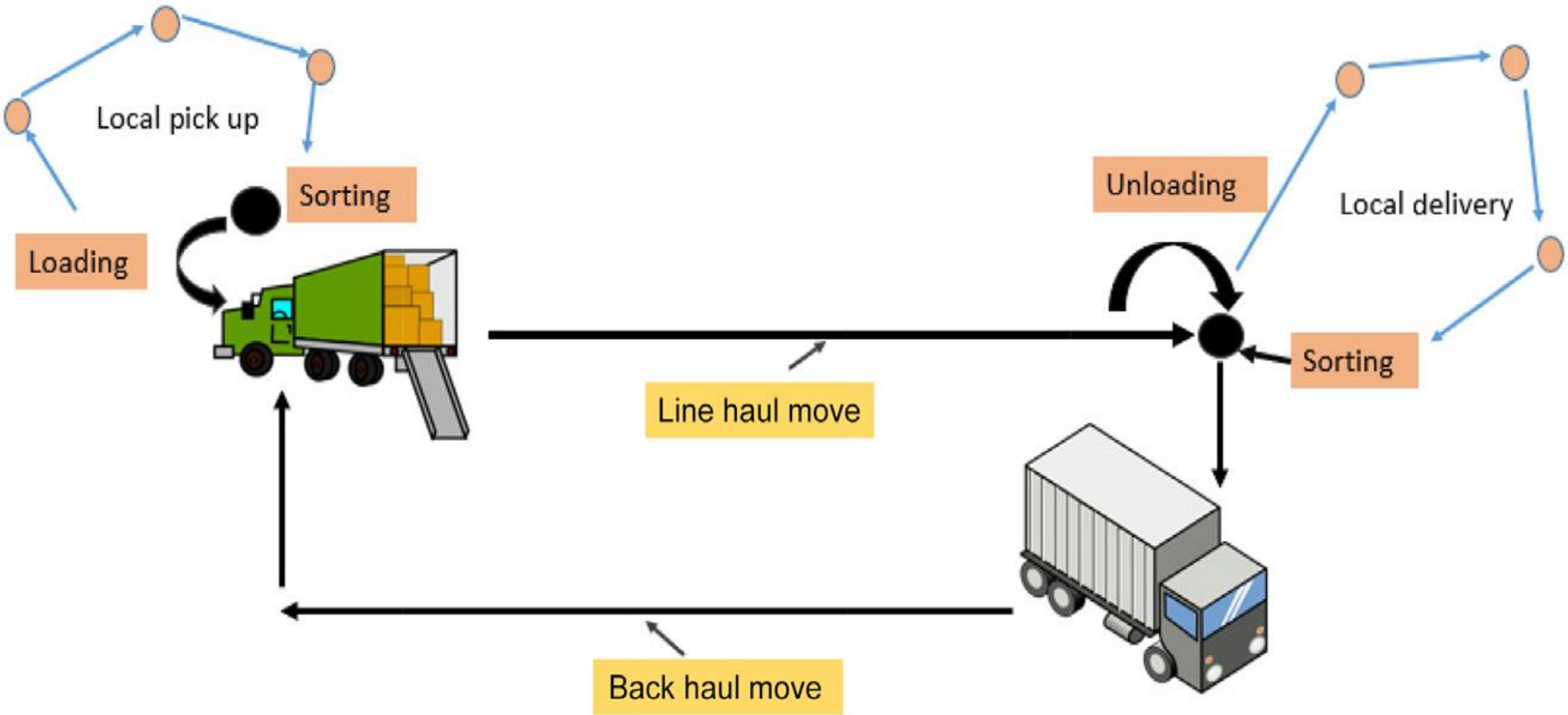
Layers of Transport network

- Transportation networks can be divided into three layers; physical network, operational network, and strategic network.
- The physical network is the actual physical path that the product takes during its shipment from origin to destination. The physical network has the following components:
- Guideways
- Terminals
- Control

Transport Operational network

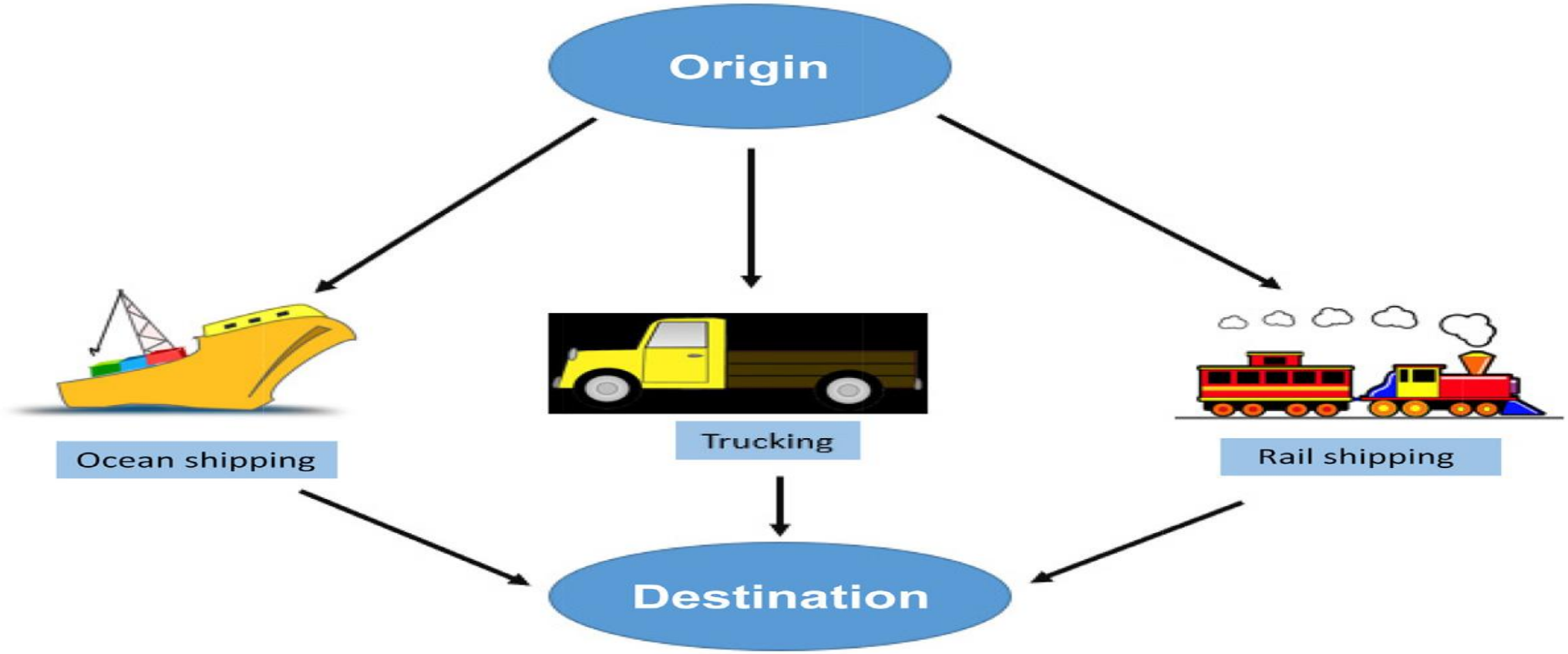
The four primary components of the operational network are loading/unloading, local routing, line haul, and sorting.




Local pick up
Loading
Sorting
Line haul move
Unloading
Local delivery
Sorting
Back haul move

Strategic transport network

- Looking at whole routes as individual links with total costs attached.
- This is a series of paths through the network from origin to destination. Each represents a complete option and has end-to-end cost, distance, and service characteristics.

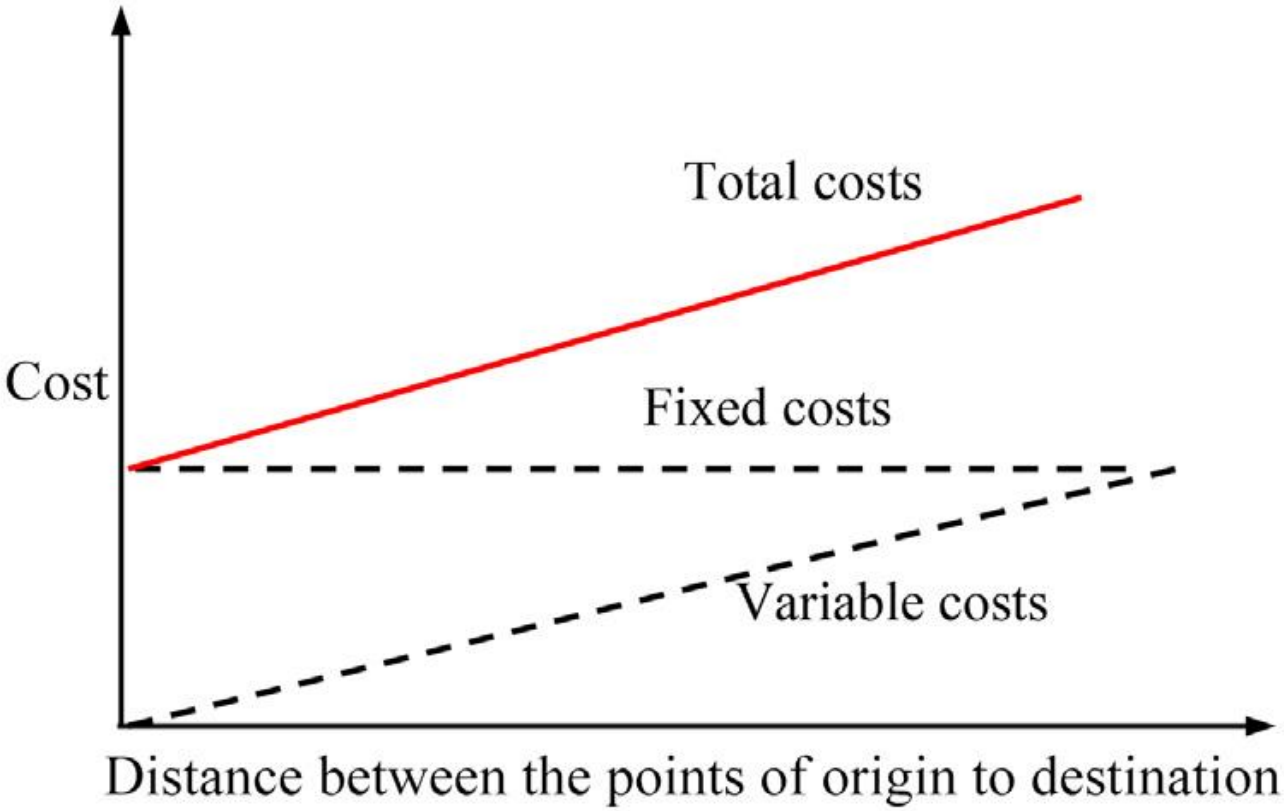


Transportation cost components

- The four basic cost components are line haul, pickup and delivery, terminal handling, and billing and collecting costs.
- Line haul: This cost component is based on the distance traveled by the delivery vehicle and the weight and space occupied by the delivery product.
- Pickup and delivery: This cost component is associated with the time spent to load or unload a transportation vehicle regardless of whether it is full or not.
- Terminal handling: This cost component depends on the number of times a shipment needs to be loaded, handled, and unloaded.
- Billing and collecting: There is cost associated with the paperwork for each shipment and generating invoices.

Transportation cost analysis

Cost per item= Holding(inventory) cost+ Moving cost



A = Fixed order cost ($/shipment)

r = Inventory holding cost ($/year)

v = Purchase cost ($/item)

Q = Shipment size (items)

Q_h = Handling size (items)

$Q_{h\max}$ = Maximum handling size (items)

D = Annual demand (items)

T = Shipment frequency (year) = Q/D

L = Lead time for transport (year)

c_f = Fixed transport cost ($/shipment)

c_v = Variable transport cost (#/item)

c_{fh} = Fixed handling cost ($/shipment)

c_{vh} = Variable handling cost (#/item)

c_s = Fixed cost per stop ($/stop)

c_d = Cost per distance ($/distance)

d = Distance traveled

c_{vd} = Marginal cost/item/distance

c_{vs} = Marginal cost/item/stop

n_s = Number of deliveries stops

Moving cost = Shipment cost + Handling cost and
Holding (inventory) cost = Inventory holding cost × Average annual inventory
$= rv(Q/2) = rv(TD/2)$
Inventory cost per item $= rv(T/2)$

Shipment(transport)or movement cost = Inventory holding cost
× Average annual inventory
Shipment cost $= c_f + c_v Q$

where $c_f = c_s(1+n_s) + c_d d c_v = c_{vs} + c_{vd} d$

$$\text{Shipment cost} = \left[c_f = c_s(1+n_s) + c_d d\right] + [Q(c_v = c_{vs} + c_{vd})d]$$

$$\text{Transport CPI} = c_s\left(\frac{1+n_s}{Q}\right) + c_d\left(\frac{d}{Q}\right) + c_{vs}$$

$$\text{Transport and handling cost} = c_s\left(\frac{1+n_s}{Q}\right) + c_d\left(\frac{d}{Q}\right) + c_{vs} + c_{vh} + \left(\frac{c_{fh}}{Q_{h\text{MAX}}}\right)$$

$$\text{Total cost per item} rv\left(\frac{T}{2}\right) + c_s\left(\frac{1+n_s}{Q}\right) + c_d\left(\frac{d}{Q}\right) + \left[c_{vs} + c_{vh} + \left(\frac{c_{fh}}{Q_{h\text{MAX}}}\right)\right]$$

Transport cost problem

Dell Computer Inc. bought $525 of materials from its vendor for a particular order of 5 computers. To deliver this particular order, the Dell driver needs to stop in 4 different places to cover a total distance of 250 miles; the cost per stop is $0.43 and cost per mile is $0.5 since the company is using UPS for delivery. The marginal cost per stop is $0.25, the fixed cost of handling a pallet of computers (pallet size55) is $10, and the variable cost of handling the same pallet is $7.50. Compute

1. How much will Dell charge its customer for shipping and handling for each computer?

2. What would be the total cost per computer if Dell holds its inventory for an average 2 weeks at $0.25/year against every $1 of inventory?

Given data

- r= $0.25/ year= 0.25/52 per week >>> for two week $0.0096
- V= Purchase cost/item = 525/5=$105
- Q= Shipment size =5
- Qh= 5
- Qhmax= 5
- Cfh= fixed handling cost= $10/shipment
- Cvh= variable handling cost per item= 7.5/5= $1.5/computer
- Cs= Fixed cost per stop=$0.43/stop
- Cd= Cost per mil=0.5/mile
- d=250 miles total distance
- Cvs=marginal cost/item/stop= 0.25/5= 0.05/computer/stop
- Ns=number of delivery stop= 4

From equation 1,

Transport and handling cost= $28.98

Total cost per computer= Inventory cost + T&H cost

rv(Q/2)+28.98

=0.0096*105*5/2+ 28.98

= $31.50

Company's warehouse to Distributors warehouse transport problem

Objective: The optimal quantity of automobile part supply chain that should be delivered from company's regional warehouse to different dealers service stations in order to minimise the transport cost.

Company Regional Warehouse	Dealers service stations
Vizag	Raipur
Vijaywada	Bhubaneshwar
Hydrabad	Vizag
	Vijayawada
	Guntur
	Rajahmundry
	Karimnagar

Average shipping cost of per packet

	Vizag	Vijaywada	Hydrabad
Vizag	15	160	100
Vijaywada	160	12	260
Raipur	154	315	56
Bhubnewar	245	410	190
Guntur	130	290	58
Rajahmundri	125	427	204
Karimnagar	215	375	160

- Storage capacity of company warehouse

Vizag>>> 6000

Vijayawada>>> 4000

Hyderabad>>> 5000

Demand at distributors warehouses

- **Vizag 1200**
- Vijayawada 1600
- Raipur 1800
- Bhubaneswar 2100
- Guntur 1300
- Rajahmundry 2900
- Karimnagar 1700

REVISED GVW OF TRANSPORT VEHICLES AFTER REVISED SAFE AXLE LOAD AND 5% TOLERANCE BY MoRTH (16-07-2018)													
SN	AXLES AND TYRES TYRES & VEHICLE'S COMMON NAME	VEHICLE WITH AXLES AND WHEELS DIAGRAM	NUMBER, POSITION AND TYRES OF AXLES								GVW (KGS)		
			FRONT				REAR			TOTAL AXLES	PRESENT	ENHANCED	With 5%. Tolerance
			1st	2nd	3rd		1st	2nd	3rd				
1	6tyre truck		2				4			2	16200	19000	19950
2	10 tyres multy axle truck		2				4	4		3	25000	28500	29925
3	12 tyre single chassis rigid truck		2	2			4	4		4	31000	36000	37800
4	12 tyre single chassis rigid truck		2				2	4	4	4	31000	36000	37800
5	14 tyre single chassis rigid truck		2	2			2	4	4	5	37000	43500	45675
6	14 tyre single chassis rigid truck		2	2			4	4	2	5	37000	43500	45675
7	14 tyres semi trailer		2	4			4	4		4	35200	40000	42000
8	18 tyres semi trailer		2	4			4	4	4	5	40200	46000	48300
9	18 tyres semi trailer		2	4	4			4	4	5	40200	49500	51975
10	22 tyres semi trailer		2	4	4		4	4	4	6	49000	55000	57750

Carrier Types

Truckload (TL) Carriers

- Specialize in moving large volumes of freight for their customers
- Target shipments that weight 15k-50k lbs
- Door-to-door service - collect at point of origin and deliver to destination without any stops
- Reduced transit times and the likelihood of delays and damage



Less-Than-Truckload (LTL) Carriers

- LTL carriers specialize in smaller loads
- Share trailer capacity among multiple shippers
- Collects freight from several shippers - bring it to a central terminal - freight is sorted based on geographic destination and them loaded onto trucks a second time (may occur multiple times)



Rail Transport

BOXNS WAGON





BOY WAGON



BRNA WAGON



BRNA –EUR WAGON



BRNA WAGON



BFNSM22.9 WAGON



BTPN WAGON



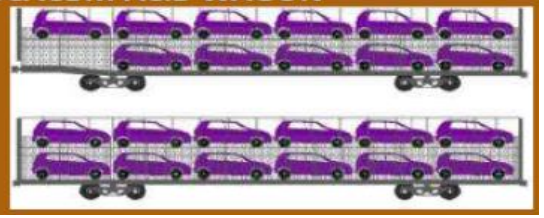
BTAP WAGON



BCACBM WAGON



BCACBM A&B WAGON



Water Transport

- Major mode for international trade – 90%
- Dominates all modes in international freight revenue
- Ships are slow, but offer tremendous capacities for volume freight, efficient fuel consumption, and low cost
- May also occur by inland lakes and rivers
- Limitation: origin and destination must be located on navigable waterways

Container Ships

- Cargo vessels where freight is stored in 20' or 40' weather tight containers
- **Twenty-foot Equivalent Units (TEU)** - volume of one 20ft container - used to measure ship capacity
- **Full container load (FCL)**
- **Less-than-container load (LCL)**



Bulk Carriers

- Merchant ships specially designed to transport **unpackaged bulk in their cargo holds** (e.g. grains, coal, ore)
- They may have one or more hold or cavities
- **Tankers -the most common type, move bulk liquid such as petroleum.**